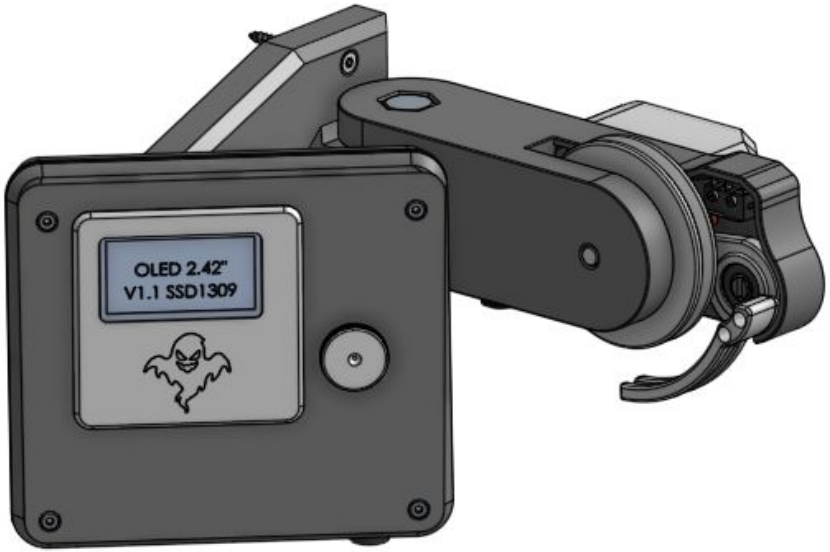
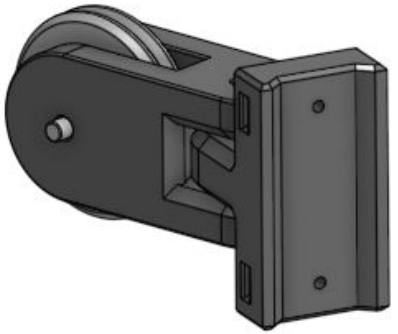
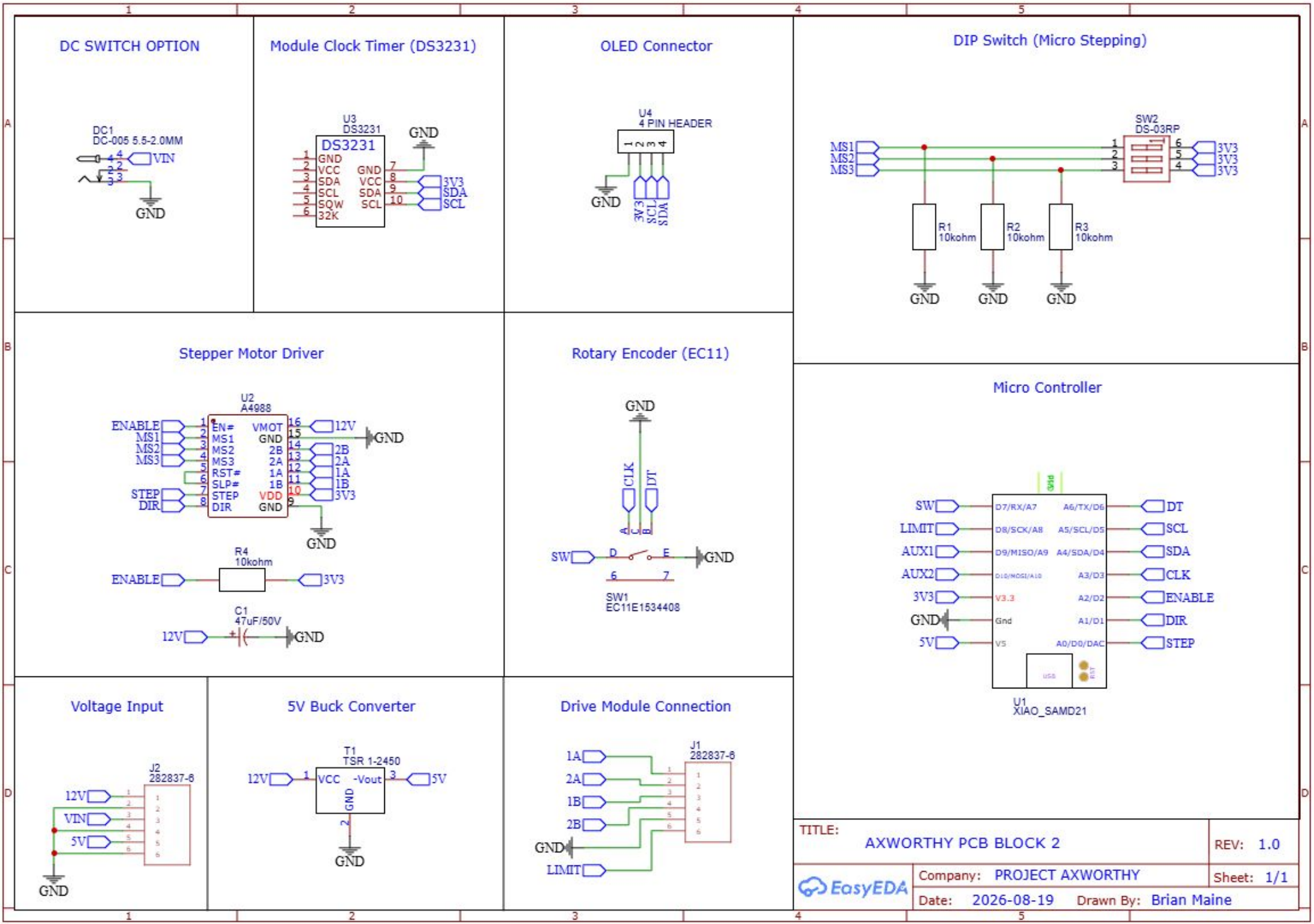
Assembly Guide

Axworthy Ghost Pulley
Rev: 3



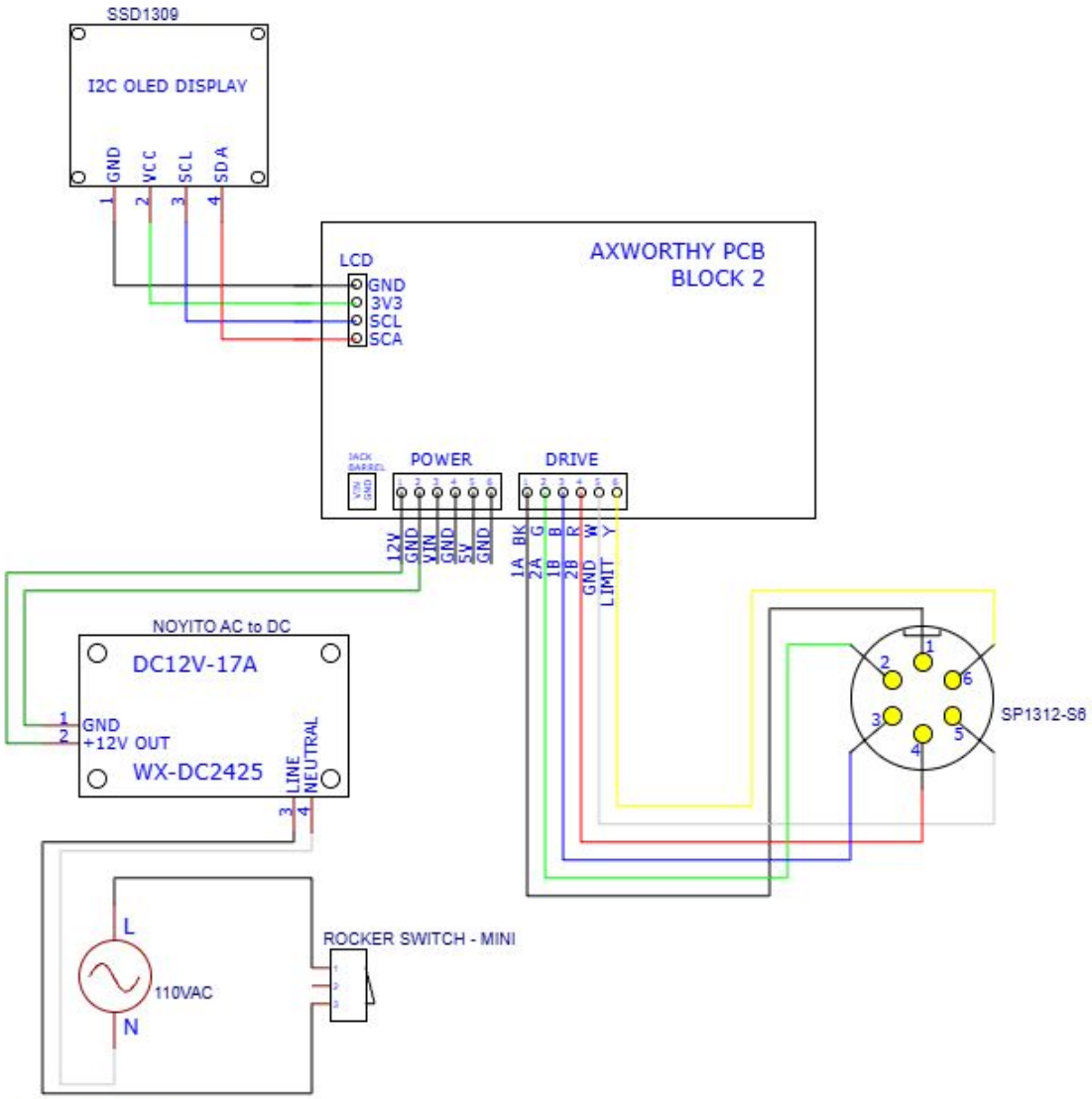


PCB Schematic



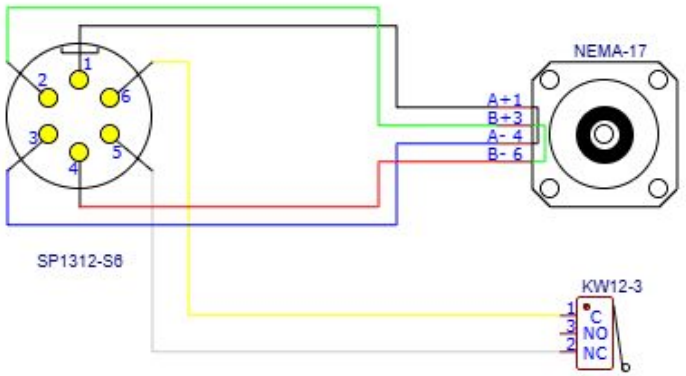
Interconnect Diagram (AC)

ENCLOSURE



CONTINUITY TESTS
Drive Terminals 1 & 3 = Continuous | Always
Drive Terminals 2 & 4 = Continuous | Always
Drive Terminals 5 & 6 = Continuous | While Limit Switch is not Pressed
Drive Terminals 5 & 6 = Not Continuous | While Limit Switch is Pressed

DRIVE HOUSING



NOTE: STEPPER PINS 1+4 AND 3+6 ARE COILS

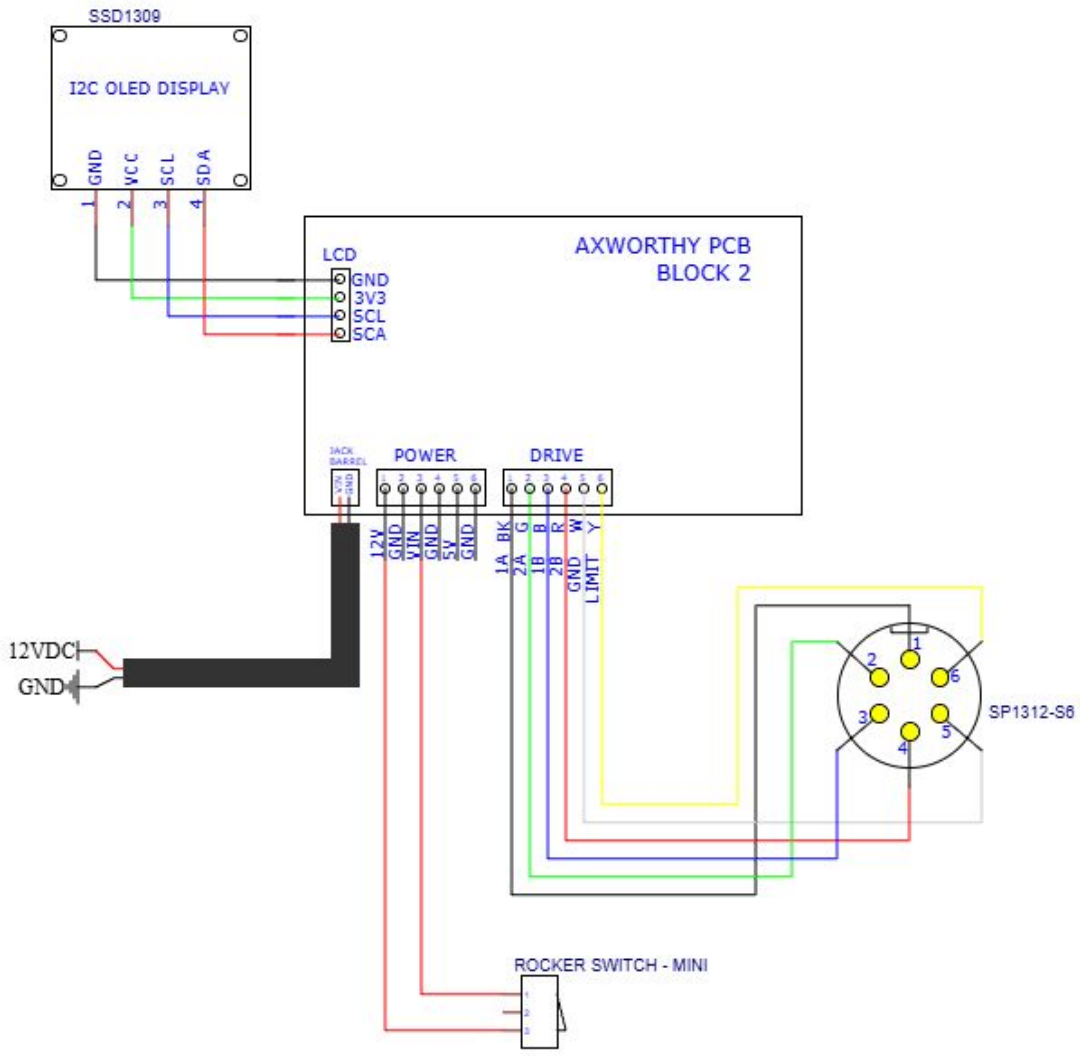
DRIVE HARNESS



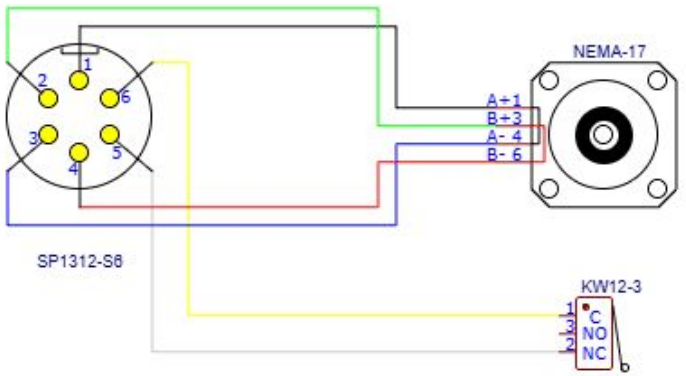
TITLE: Interconnect Diagram (AC)		REV: 2.0
EasyEDA	Company: PROJECT AXWORTHY	Sheet: 1/1
	Date: 2026-09-16 Drawn By: Brian Maine	

Interconnect Diagram (DC) - Optional

ENCLOSURE



DRIVE HOUSING



NOTE: STEPPER PINS 1+4 AND 3+6 ARE COILS

DRIVE HARNESS

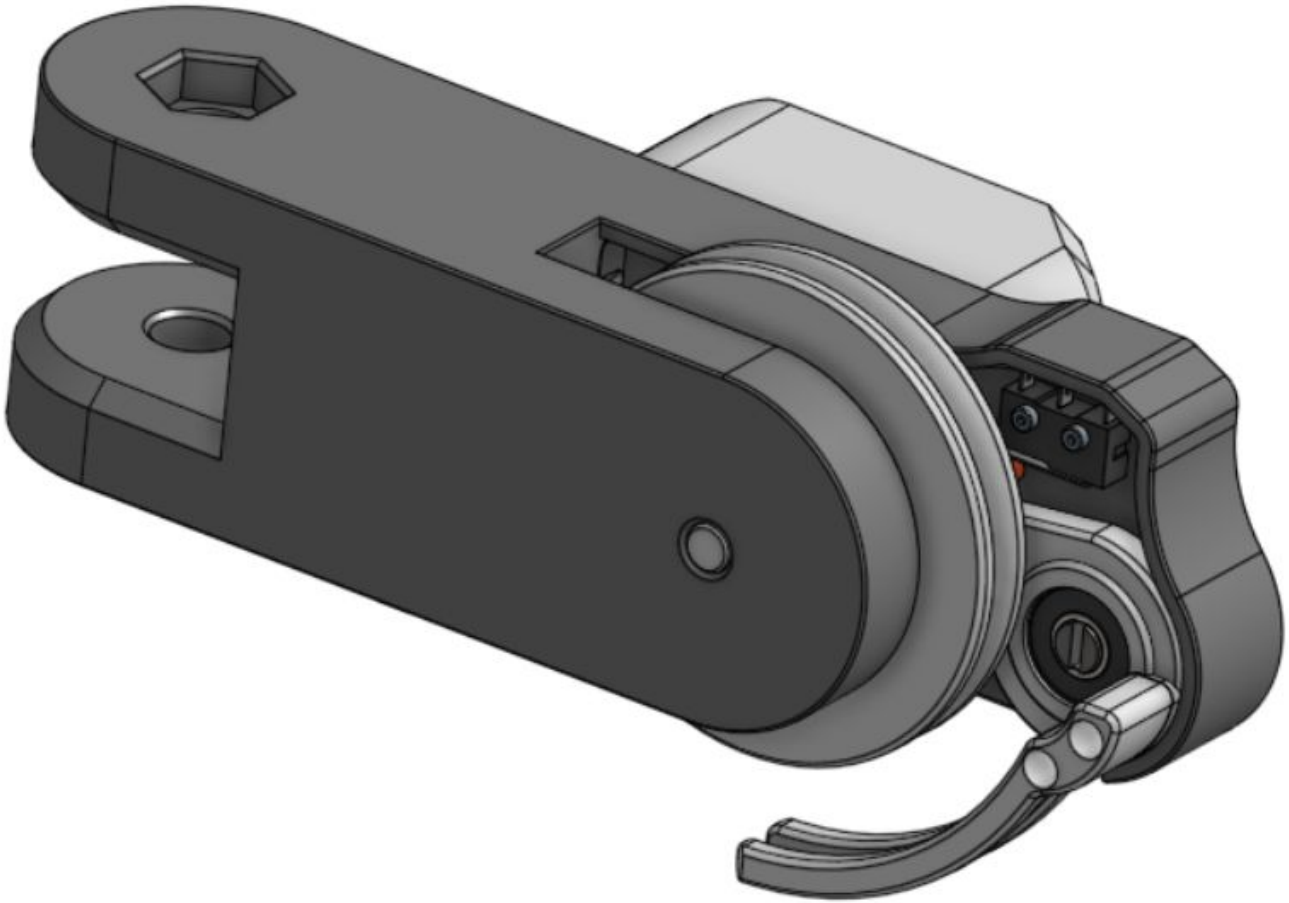


CONTINUITY TESTS
Drive Terminals 1 & 3 = Continuous | Always
Drive Terminals 2 & 4 = Continuous | Always
Drive Terminals 5 & 6 = Continuous | While Limit Switch is not Pressed
Drive Terminals 5 & 6 = Not Continuous | While Limit Switch is Pressed

TITLE: Interconnect Diagram (DC) - Optional		REV: 2.0
EasyEDA	Company: PROJECT AXWORTHY	Sheet: 1/1
	Date: 2026-09-16 Drawn By: Brian Maine	

Drive

Mechanical + Harnessing



Drive Housing

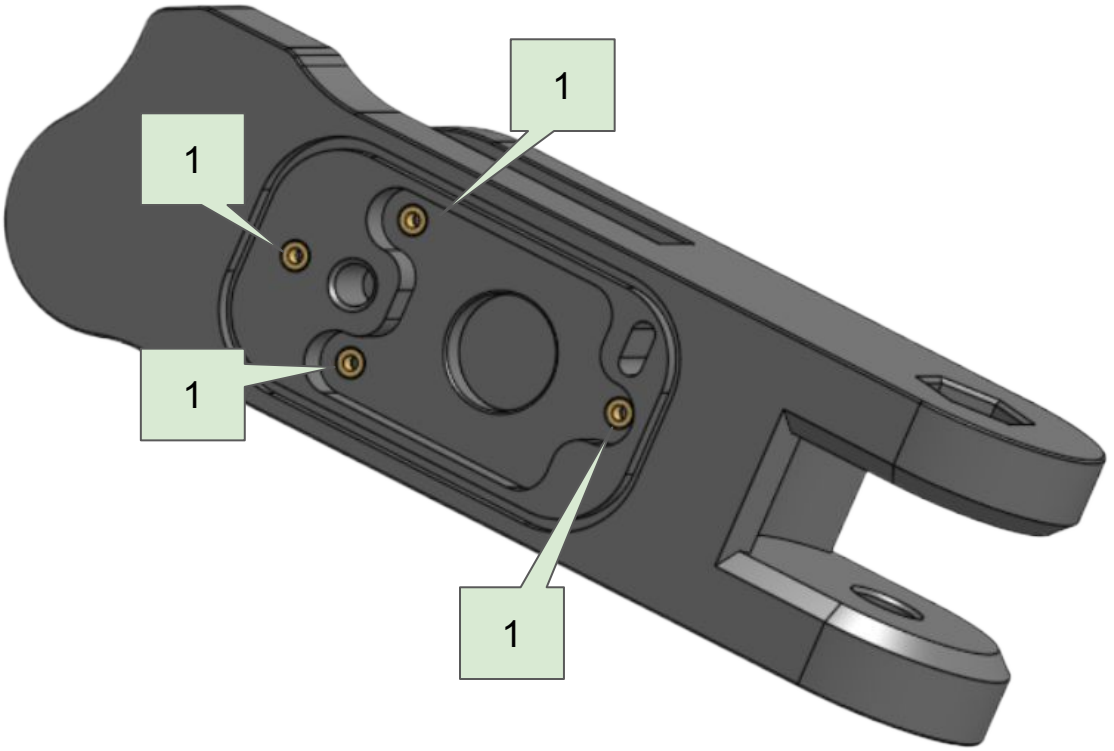
Install Hardware

Install the following Heat-set Inserts:

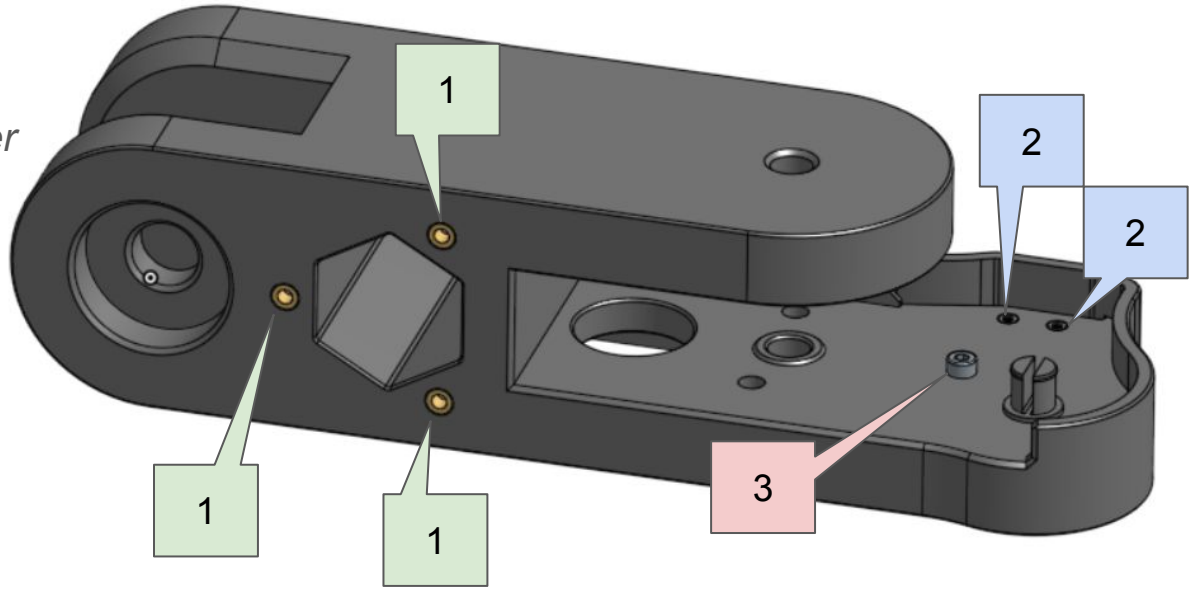
1. Heat-set insert, M3 [94180A331] [QTY. 6]
2. Heat-set insert, M2 [94180A307] [QTY. 2]

Install the following Fasteners

3. SHCS, M3x0.50, 6mm [QTY. 1]



NOTE: The SHCS in step 3 is self tapped into plastic, there is no heatsert. The head of this fastener will be used for the limit lever in a later step.



Pulley & Limit Lever A

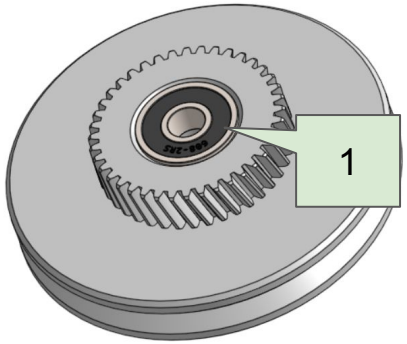
Press fit the following Bearings:

1. 608 2RS ball bearing [QTY 3]

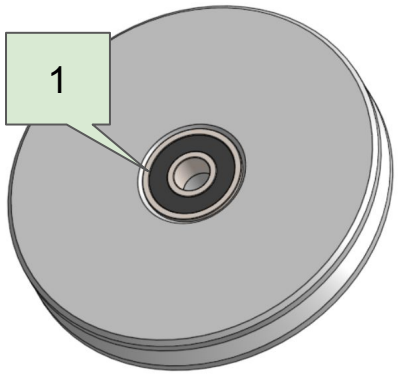
Install the following Fasteners

2. Heat-set insert, M3 [94180A331] [QTY. 2]

Drive Pulley

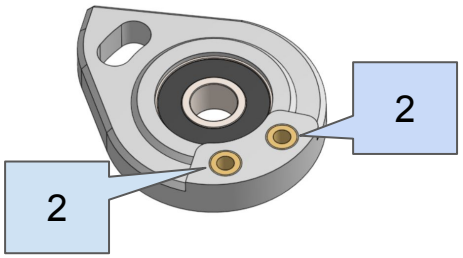


Front

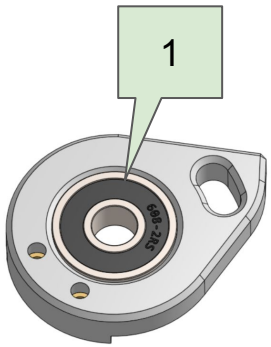


Back

Limit Lever A



Front



Back

Stepper Subassembly

Install the following Heat-set Inserts:

1. Heat-set insert, M2 [94180A307] [QTY.1]

Press Pinion onto Stepper motor shaft and install the following hardware:

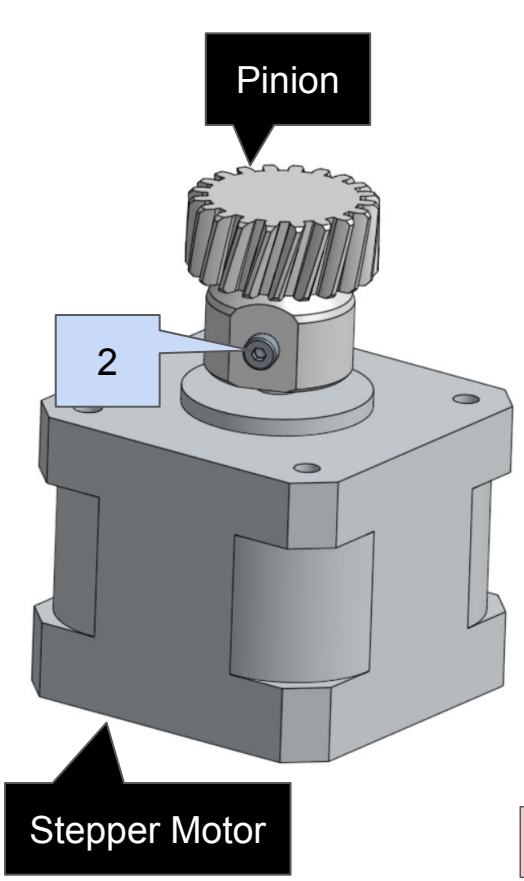
2. SHCS, M2x0.40, 6mm [QTY. 1]

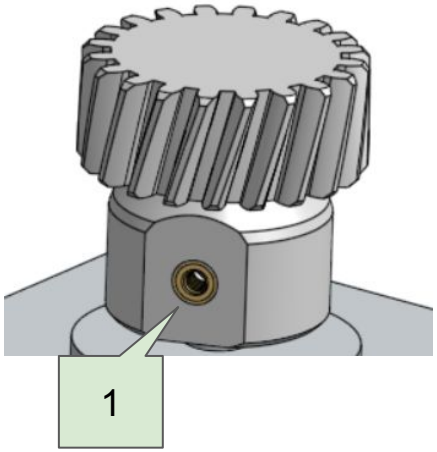
NOTE: Align Heat-sert with flat face of Stepper Motor Shaft. This ensures the pinion won't slip.

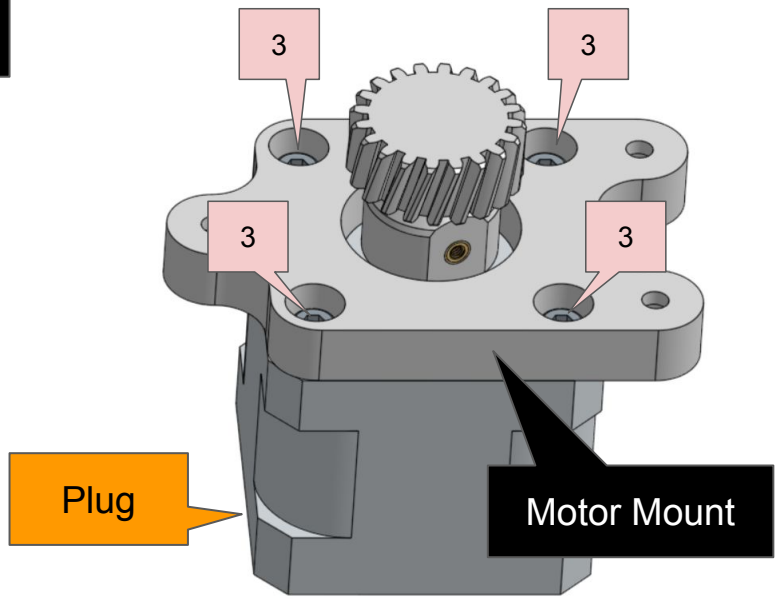
Install Motor Mount to Stepper Motor with the following Hardware:

3. SHCS, M3x0.50, 6mm [QTY. 4]

NOTE: Align plug end of stepper motor as shown in the figure

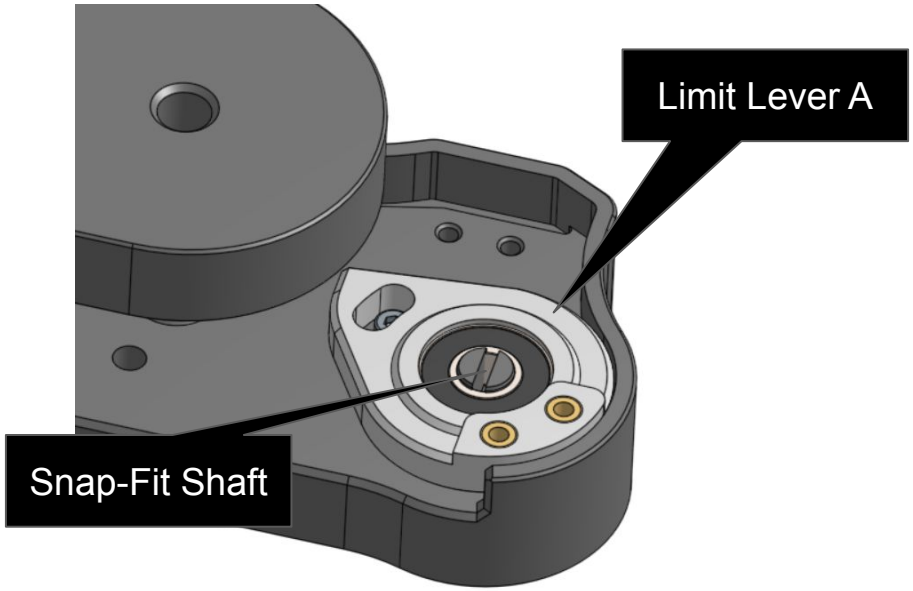






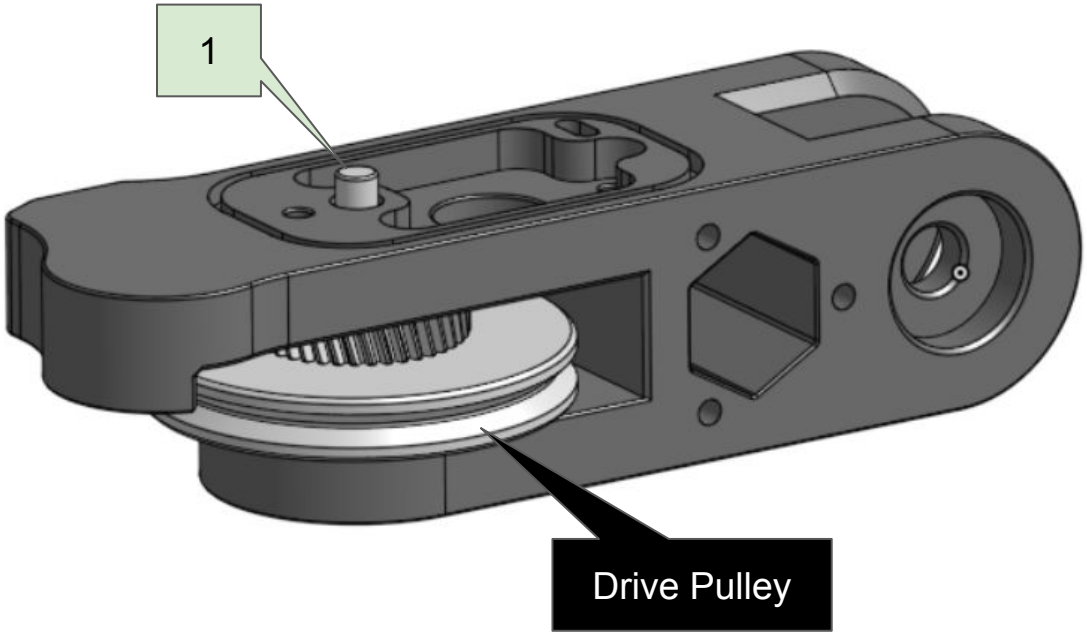
Install Drive Pulley & Limit Lever A

Install Limit Lever A into Drive Housing by pressing bearing into snap-fit shaft.



Install Drive Pulley with the following hardware:

1. Dowel pin, 8x60mm, stainless [QTY 1]



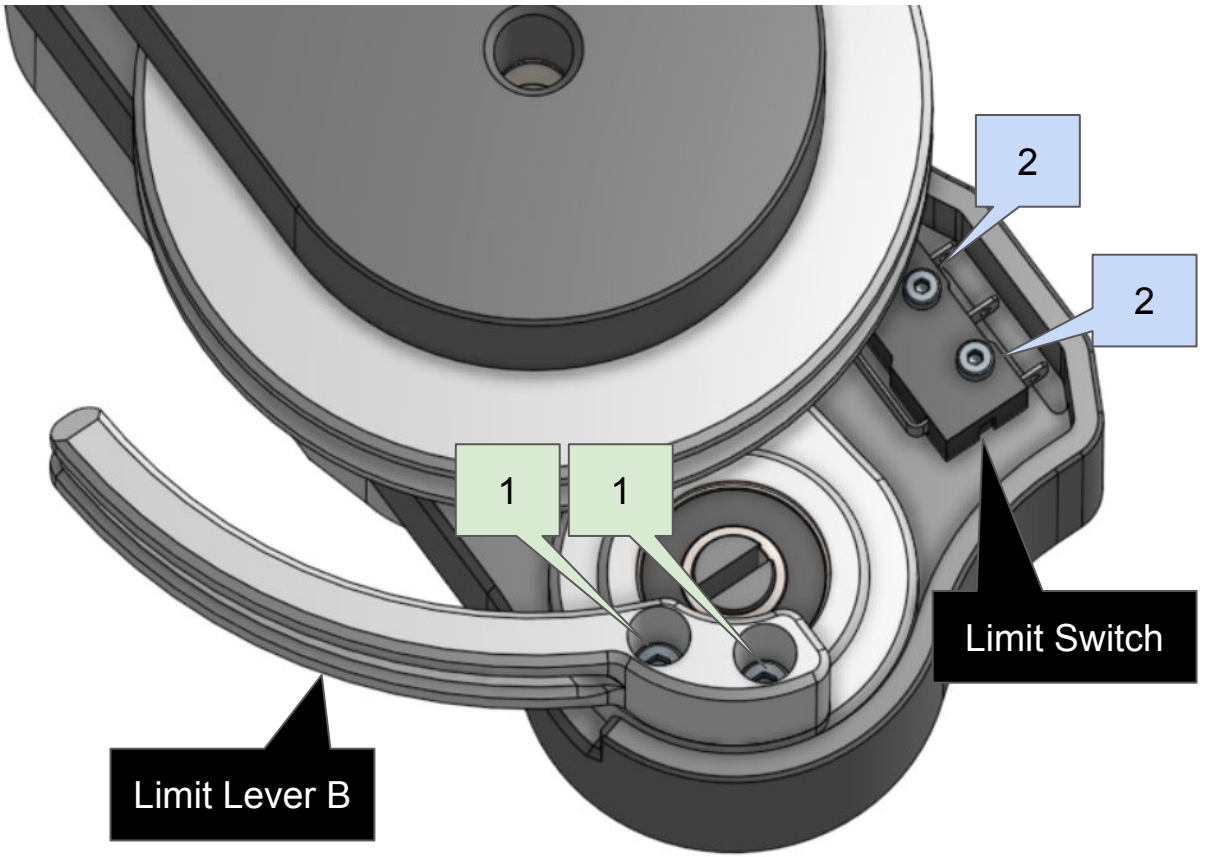
Install Limit Lever B & Limit Switch

Install Limit Lever B using the following hardware:

1. SHCS, M3x0.50, 6mm [QTY. 2]

Install Limit switch with the following hardware:

2. SHCS, M2x0.40, 10mm [QTY. 2]

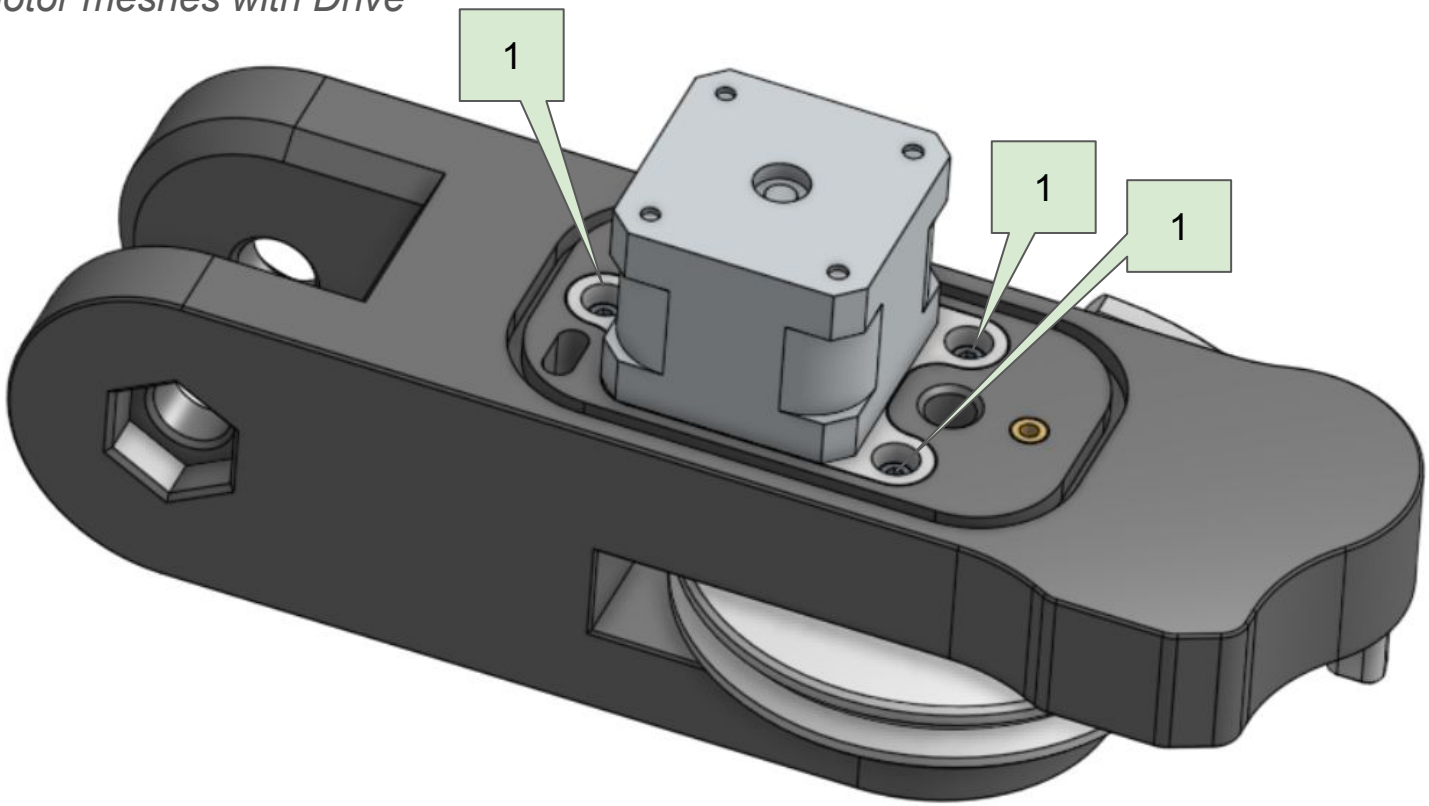


Install Stepper Subassembly

Install Stepper Subassembly with the following Hardware:

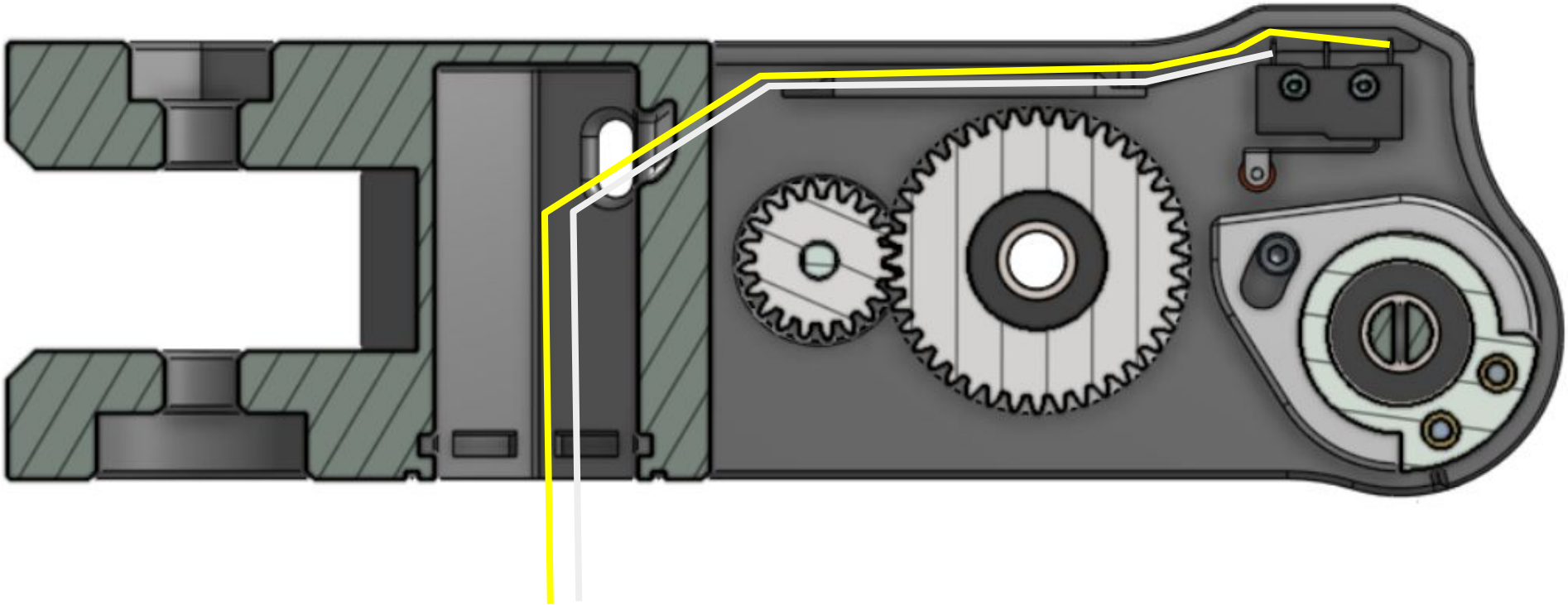
1. SHCS, M3x0.50, 6mm [QTY. 3]

NOTE: Ensure Stepper motor meshes with Drive Pulley Gear



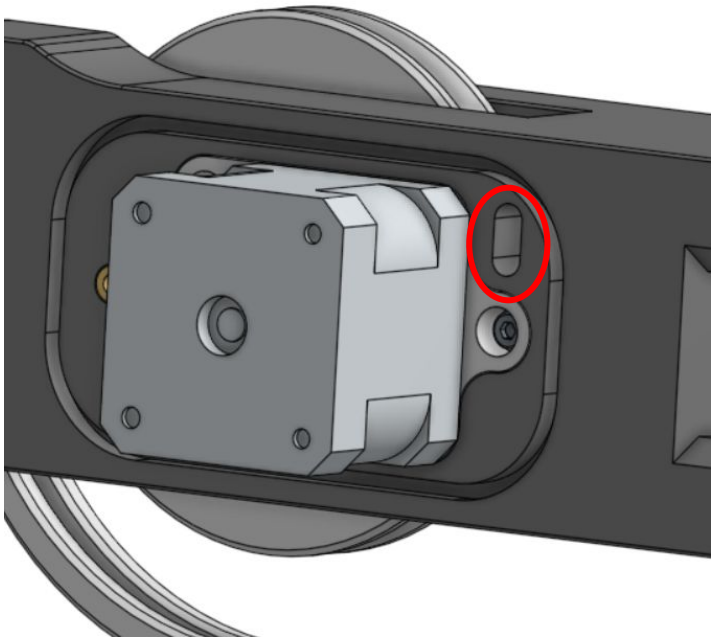
Route Limit Switch Cables

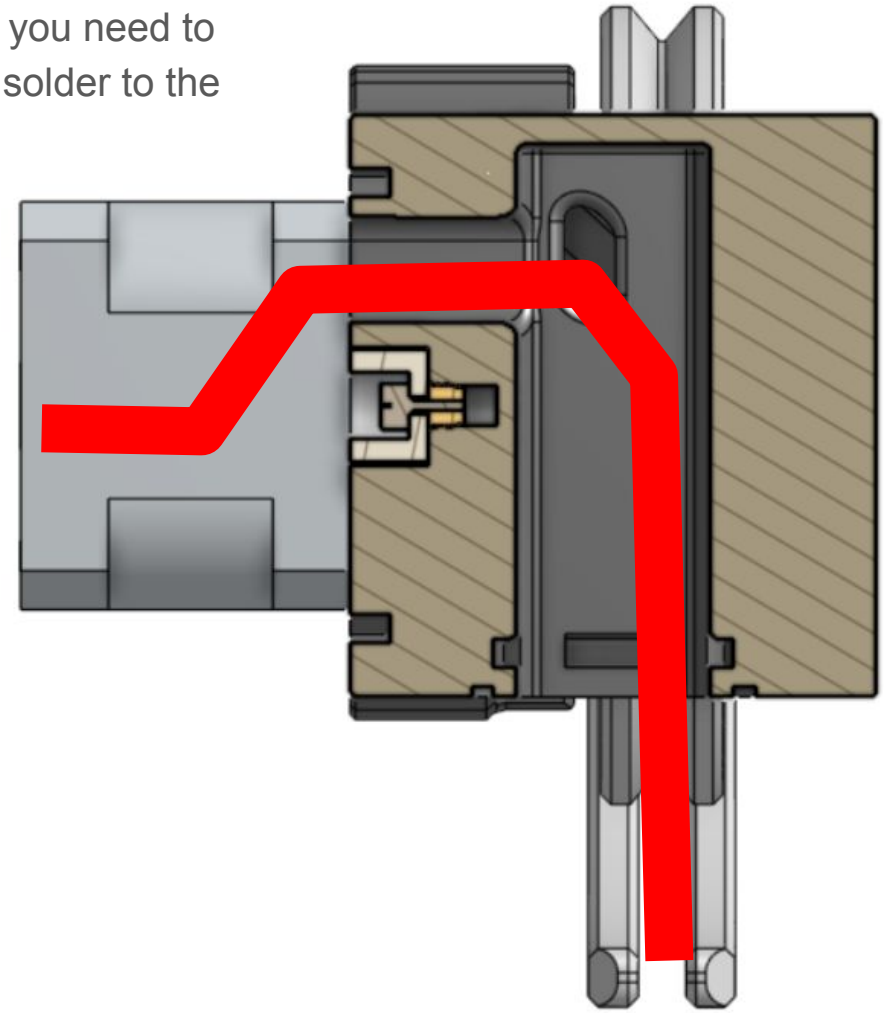
Run the Yellow and White Limit switch cables from the switch down. There is an cable tray integrated into the drive housing. Use this to route both cables through the hex cavity and out the bottom. Leave enough length to solder to the connector in a later step.



Route Stepper Cables

Route All stepper motor cables through the Drive Housing hole on the side next to the motor. You will need to cut one end of the motor cable off to fit. Cut the opposite end of the motor cable off after estimating the length you need to extend past the hex hole. Again, you'll need enough length to solder to the connector in a later step.





Install Hex Cap & Connector

Fasten the WEIPU SP13 6-pin kit SP1312 to the Hex Cap as shown.

NOTE:

- *Notice that the plug faces the chamfered edge of the hex plate*
- *Don't be alarmed of the image showing a connector with 3 pins. Yours should have 6.*

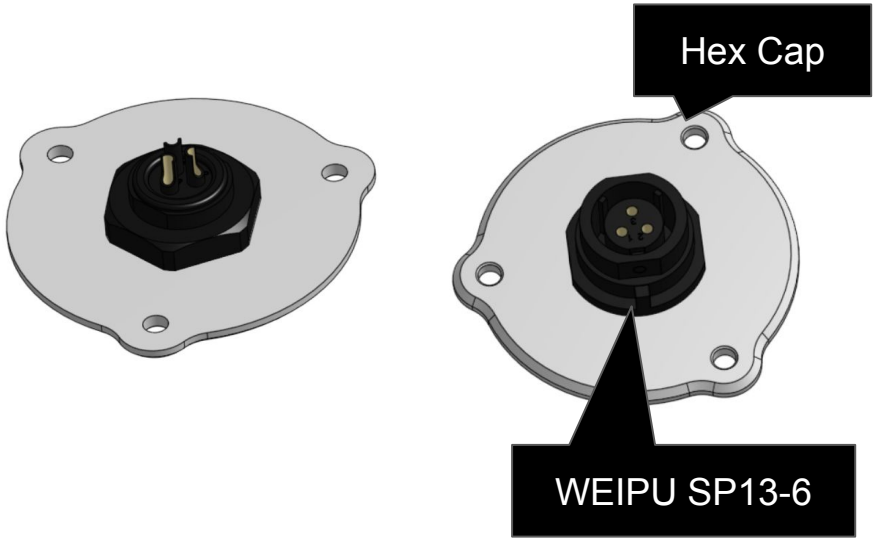
Solder Motor and Limit Switch Cables per the Interconnect Diagram (AC or DC)

Tuck the cables into the hex cavity of the Drive Housing

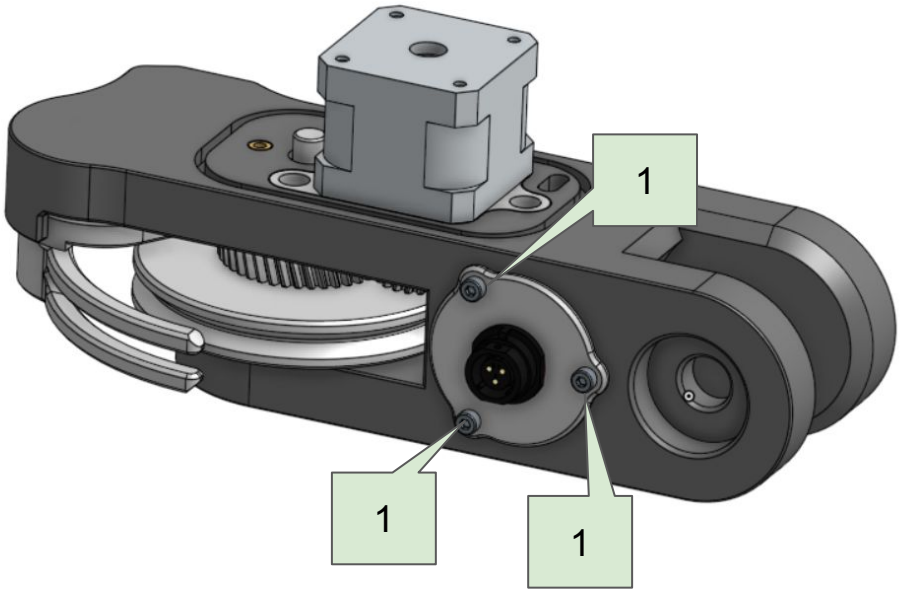
Install Hex Cap assembly to Drive Housing using the following hardware:

1. SHCS, M3x0.50, 6mm [QTY. 3]

Hex Cap Assembly



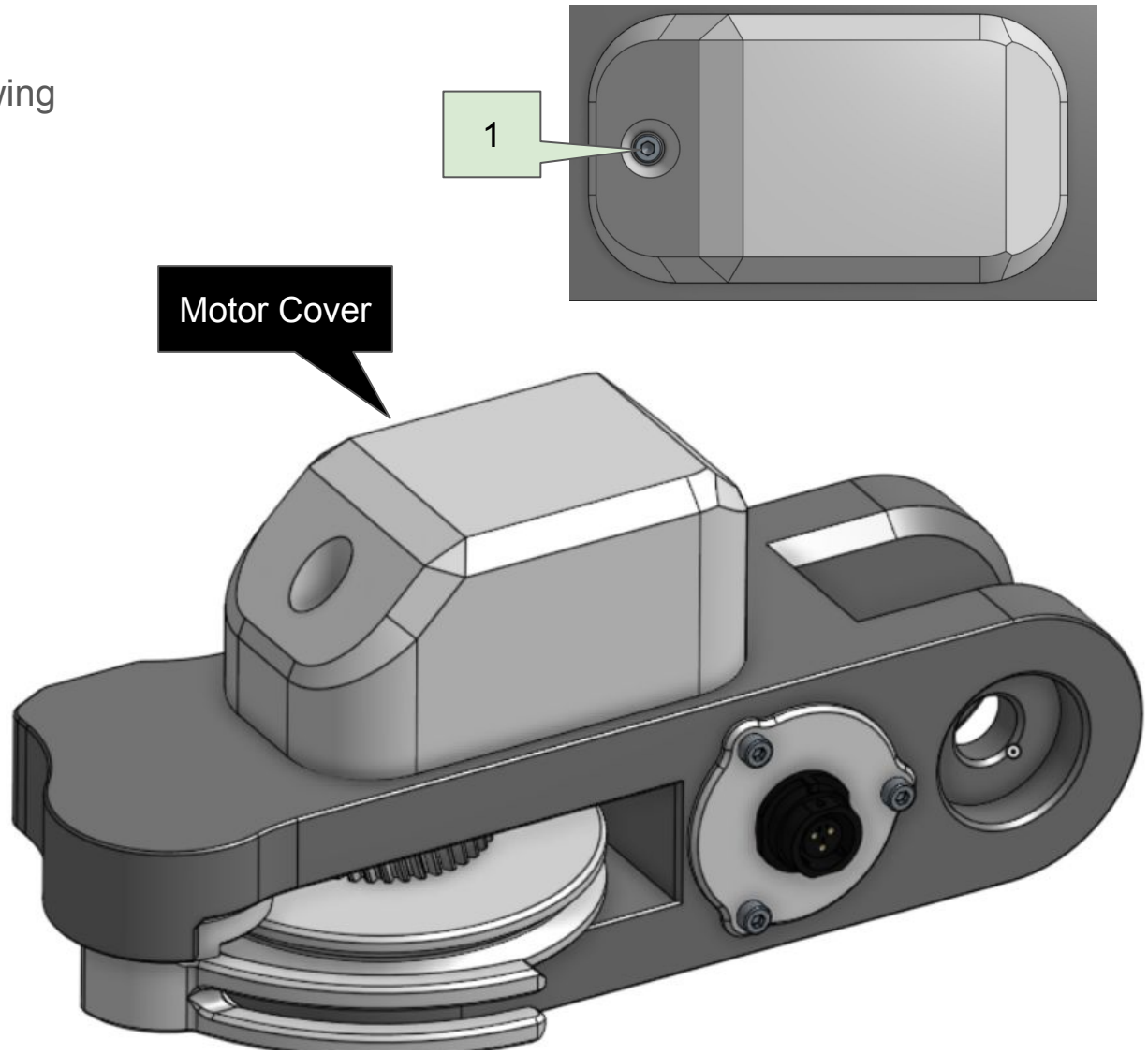
Drive Housing



Install Motor Cover

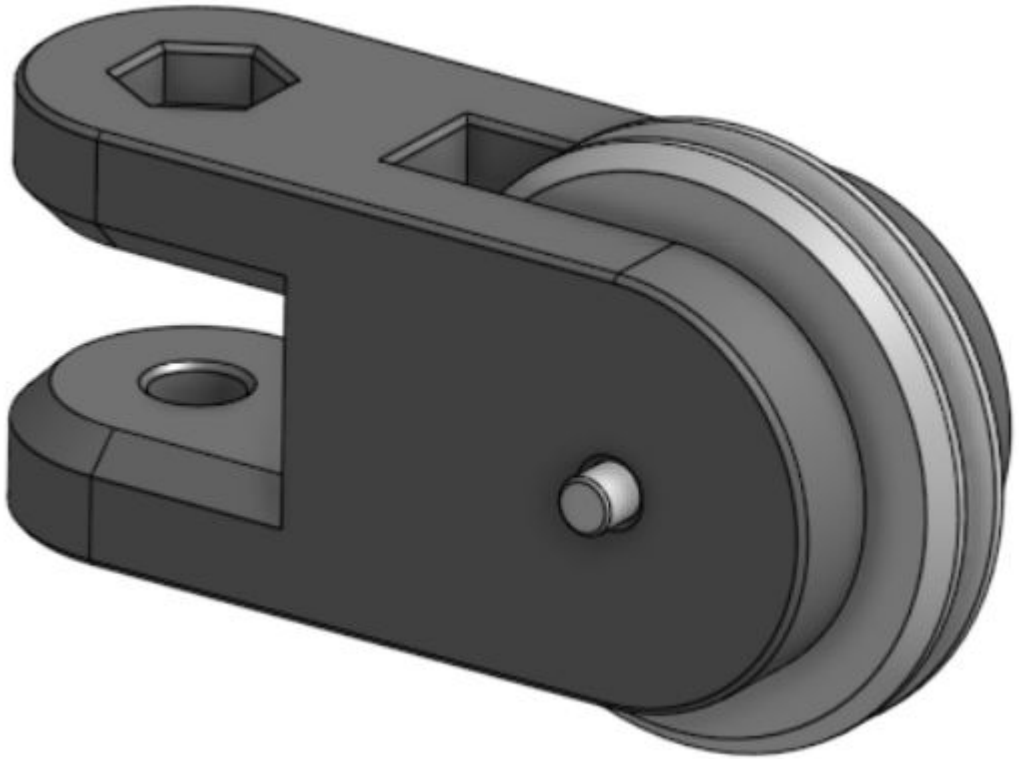
Fasten the Motor Cover with the following Hardware:

1. SHCS, M3x0.50, 6mm [QTY. 1]



Idler

Mechanical



Assemble Idler Pulley

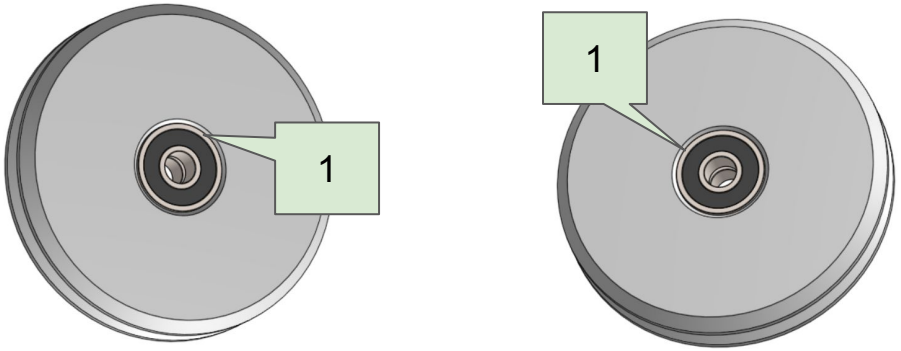
Press fit the following Bearings:

1. 608 2RS ball bearing [QTY 2]

Install Idler Pulley with the following hardware:

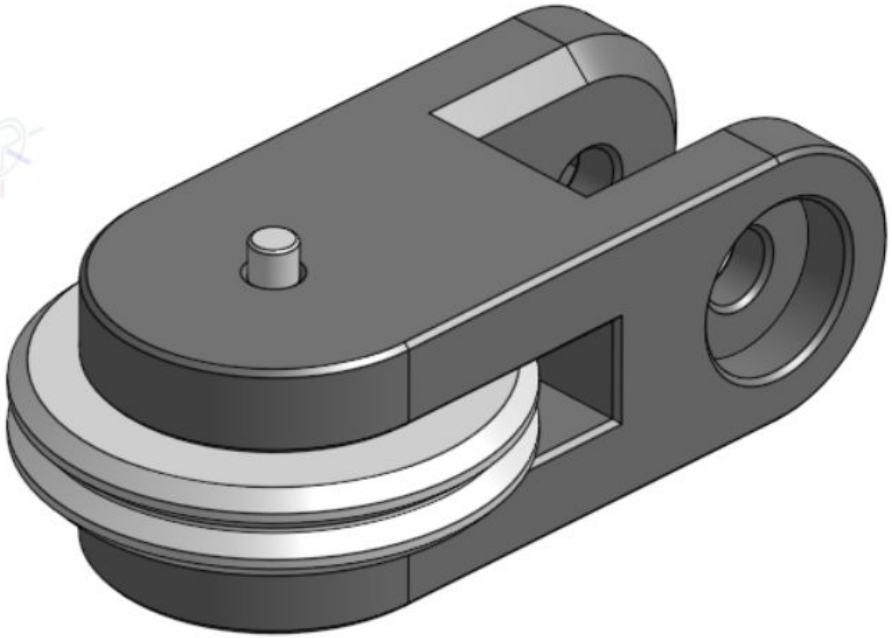
2. Dowel pin, 8x60mm, stainless [QTY 1]

Idler Pulley



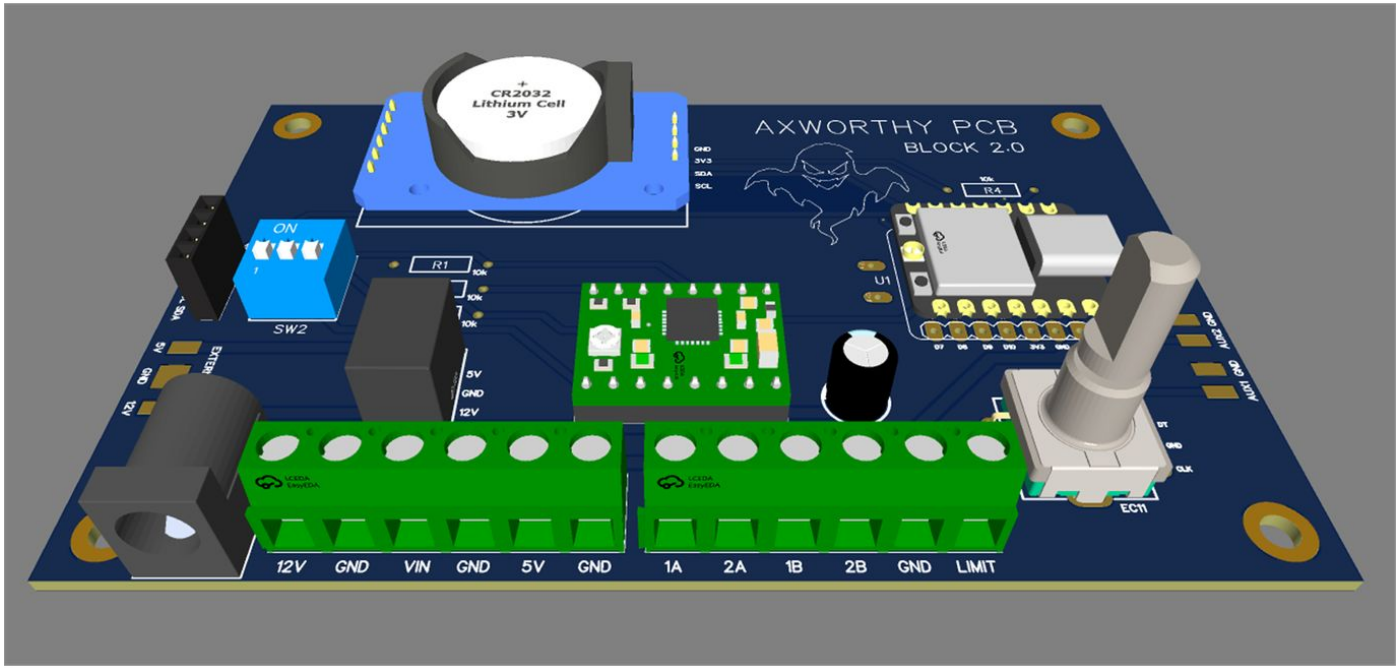
Front Back

Idler Housing



PCB

Electrical Assembly



Clock Mount

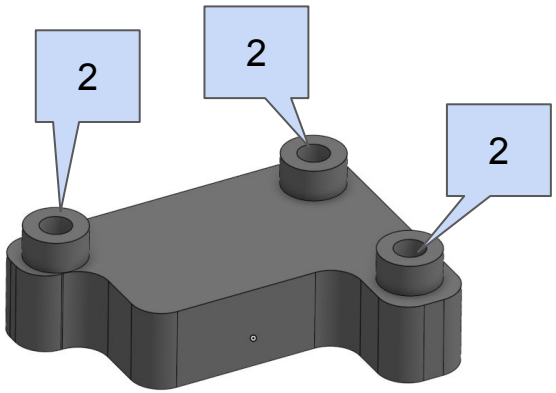
Install the following Heat-set Inserts into the Clock Mount:

1. Heat-set insert, M3 [94180A331] [QTY. 2]
2. Heat-set insert, M2 [94180A307] [QTY. 3]

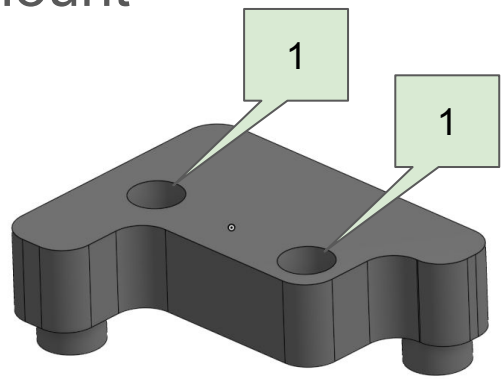
Install Clock Mount to Axworthy PCB with the following Hardware:

3. SHCS, M3x0.50, 6mm [QTY. 2]

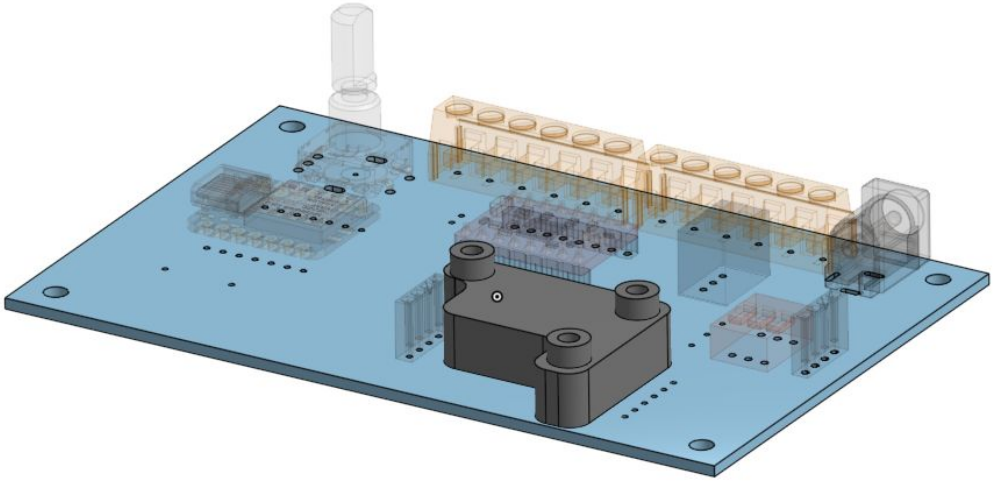
Clock Mount

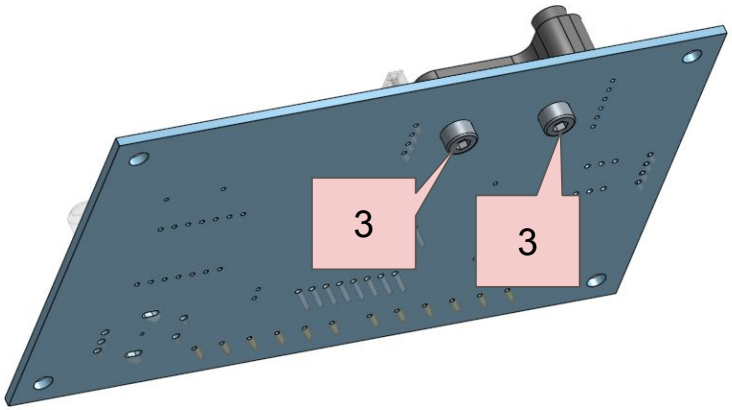


Front



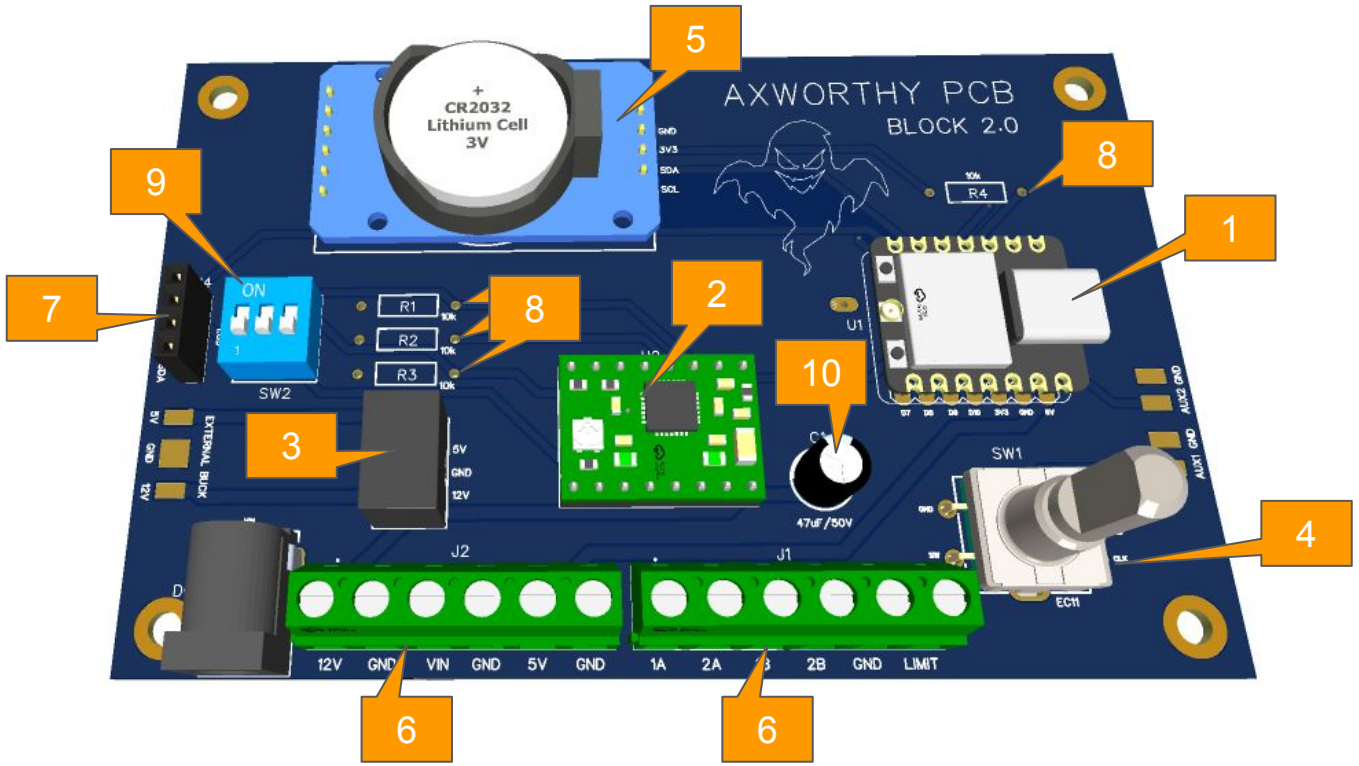
Back

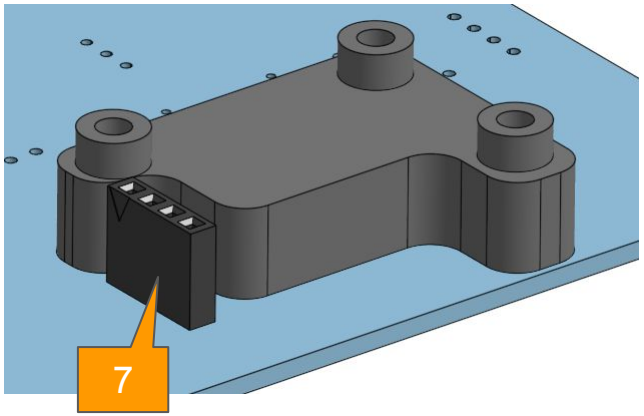




Solder PCB Components

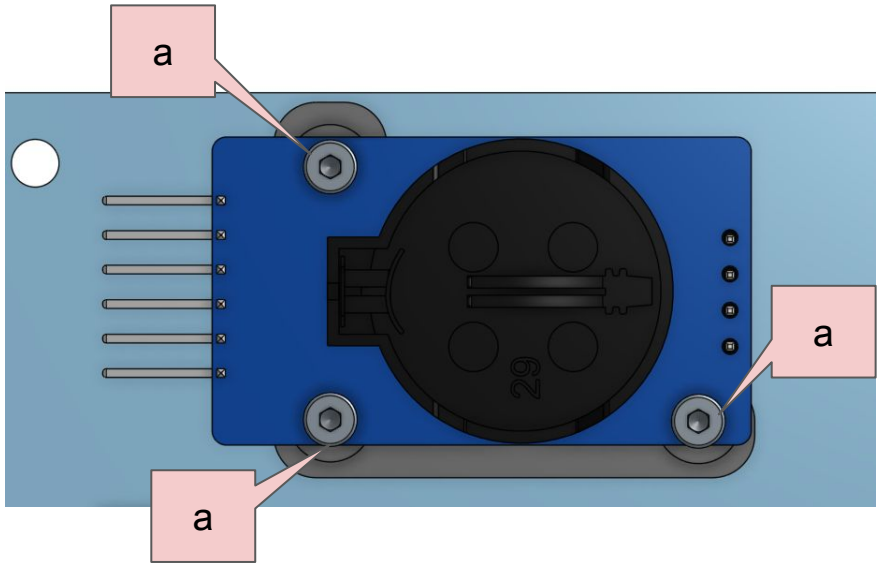
1. Seeeduino XIAO SAMD21 microcontroller, ATSAMD21G18A [QTY. 1]
2. A4988 stepper driver module [QTY. 1]
3. TSR 1-2450 Traco 12V→5V buck regulator (SIP3) [QTY. 1]
4. EC11 rotary encoder with switch [QTY. 1]
5. DS3231 RTC module [QTY. 1] (**NOT SOLDERED, Installed Later**)
6. Terminal block, 3-pin, 5mm PCB mount, 250V 8A [QTY. 4]
7. Female pin headers, 2.54mm (sockets for OLED / DS3231) [QTY. 2]
8. 10k ohm resistor (3× MS pull-down + 1× ENABLE pull-up) [QTY. 4]
9. DIP switch, 3-position, 2.54mm (MS1/MS2/MS3) [QTY. 1]
10. 47uF 50V electrolytic capacitor (A4988 VMOT bulk) [QTY. 1]
11. DC-005 barrel jack, 5.5×2.1mm, center-positive [QTY. 1]

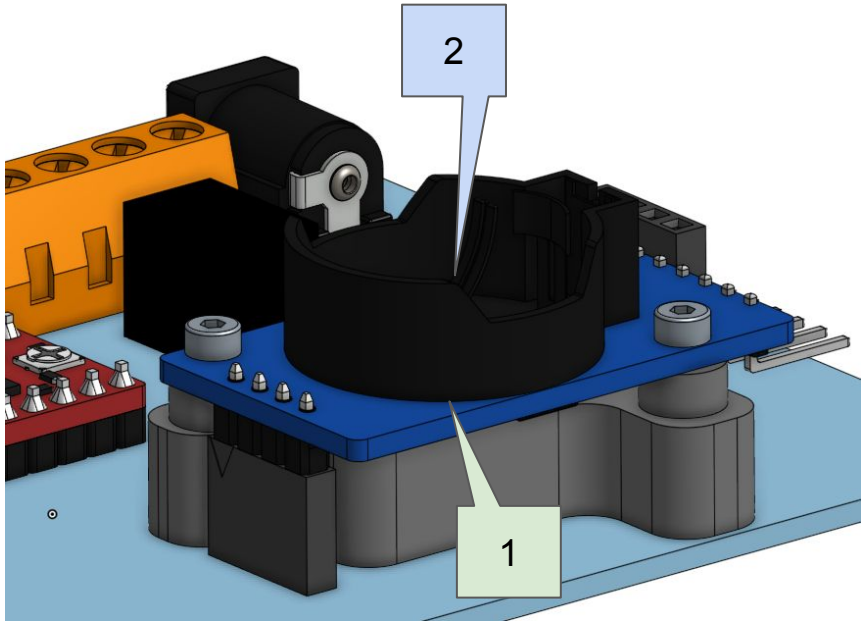




Install Clock

1. Install DS3231 RTC module [QTY. 1] using the following Hardware:
   a. SHCS, M2x0.40, 6mm [QTY. 3]
2. Install CR2032 Battery into DS3231 RTC Module





Control Box

Mechanical/Electrical Assembly

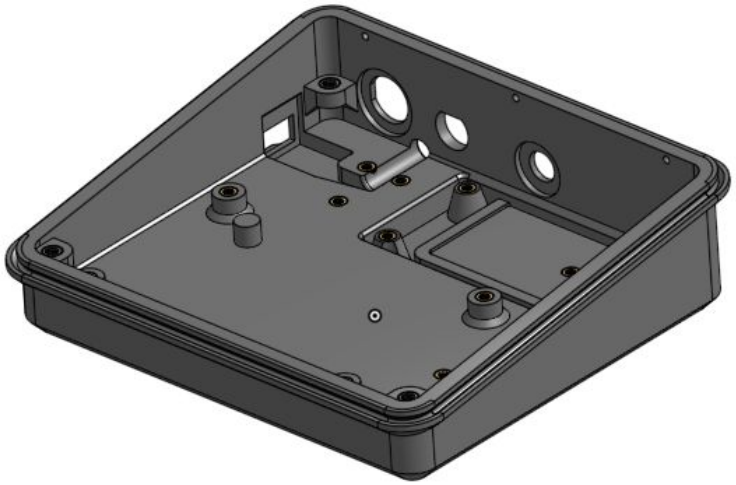


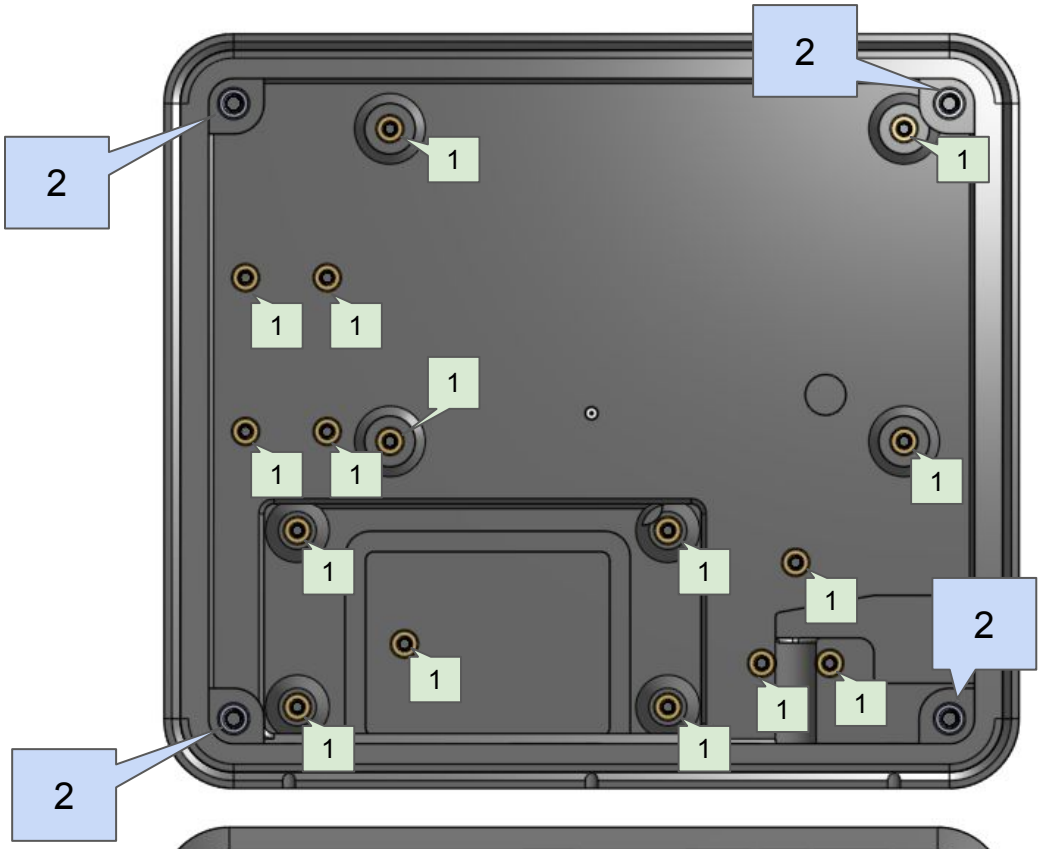
Install Heat-set Inserts

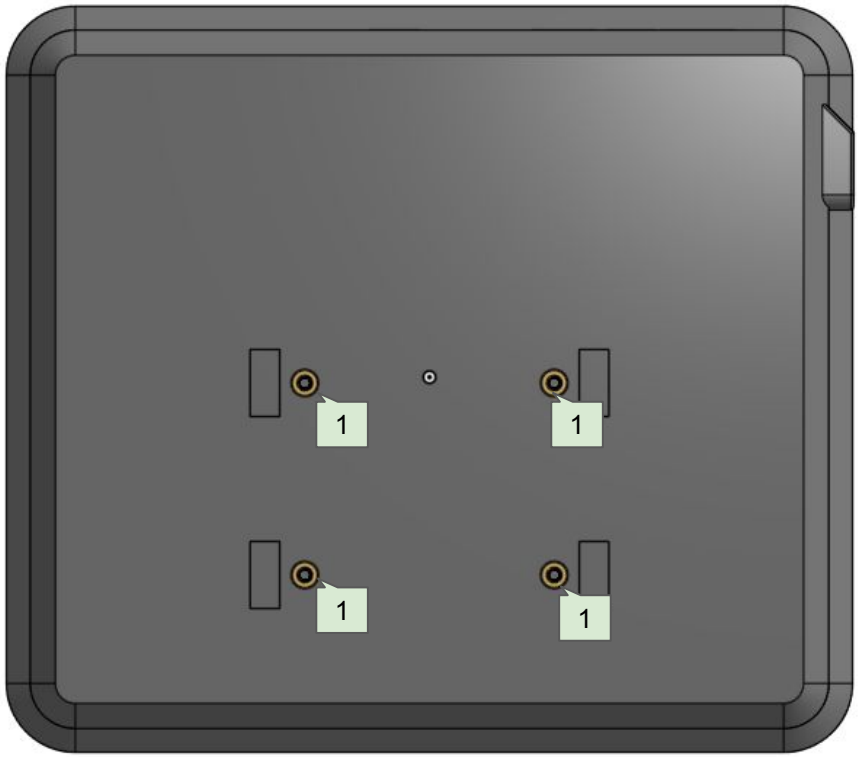
Install the following Heat-set Inserts:

1. Heat-set insert, M3 [94180A331] [QTY. 20]
2. Heat-set insert, M4 [94180A351] [QTY. 4]

Control Box





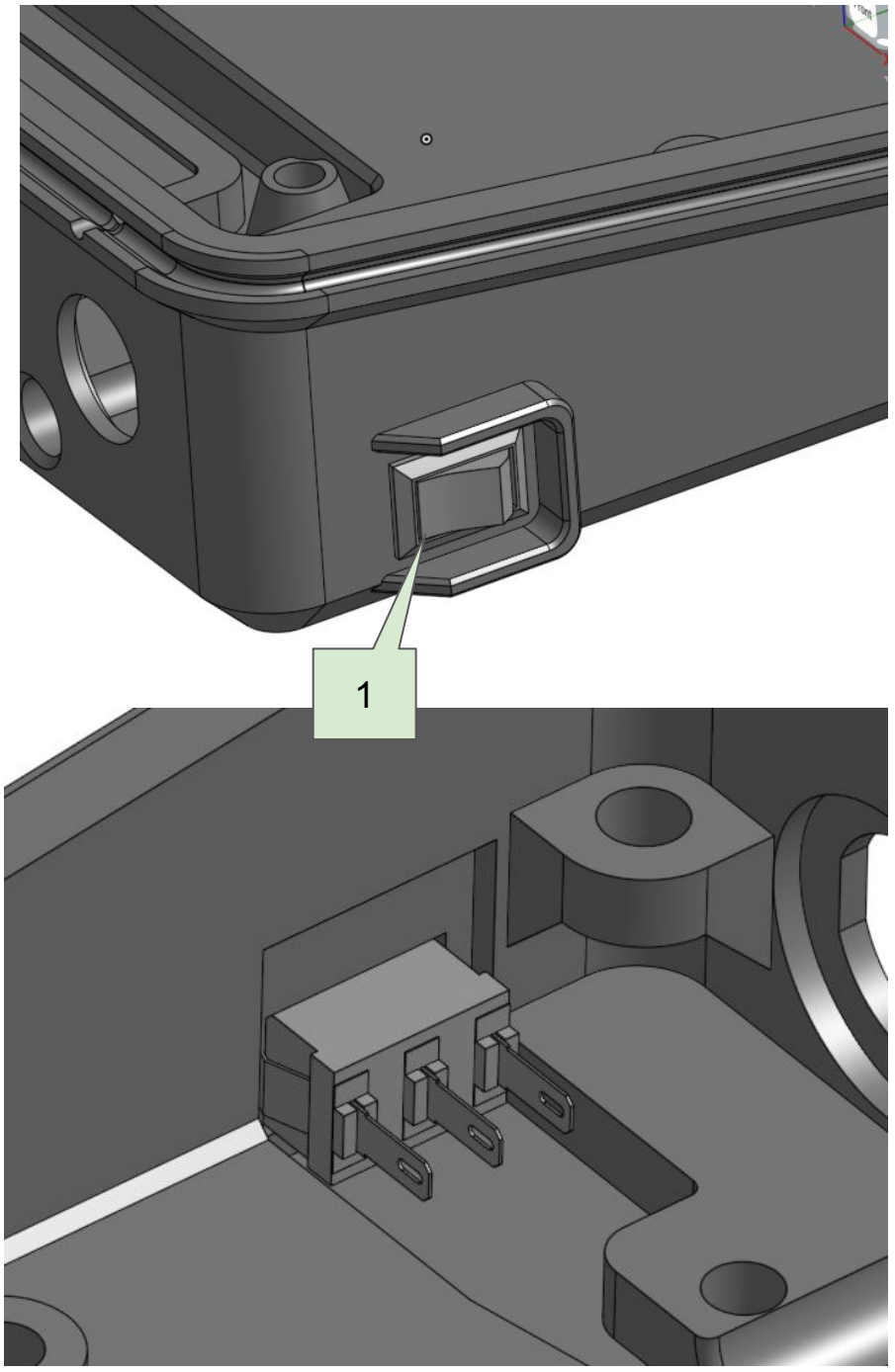


Install Rocker Switch

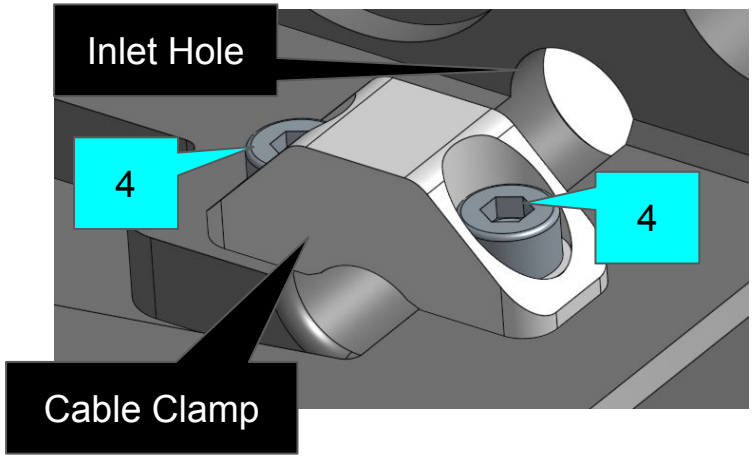
Install the following component:

1. Mini rocker switch, SPST on/off (KCD11, 10x15mm)

NOTE: Position so the [On] Position is facing up.



Box Power (AC Option)



Install The following Component and hardware

1. NOYITO AC-DC 120V->12V 2A
2. SHCS, M3x0.50, 6mm [QTY. 4]

Insert The following into the power inlet hole:

3. AC power cord (18/3 SJT, NEMA 5-15)

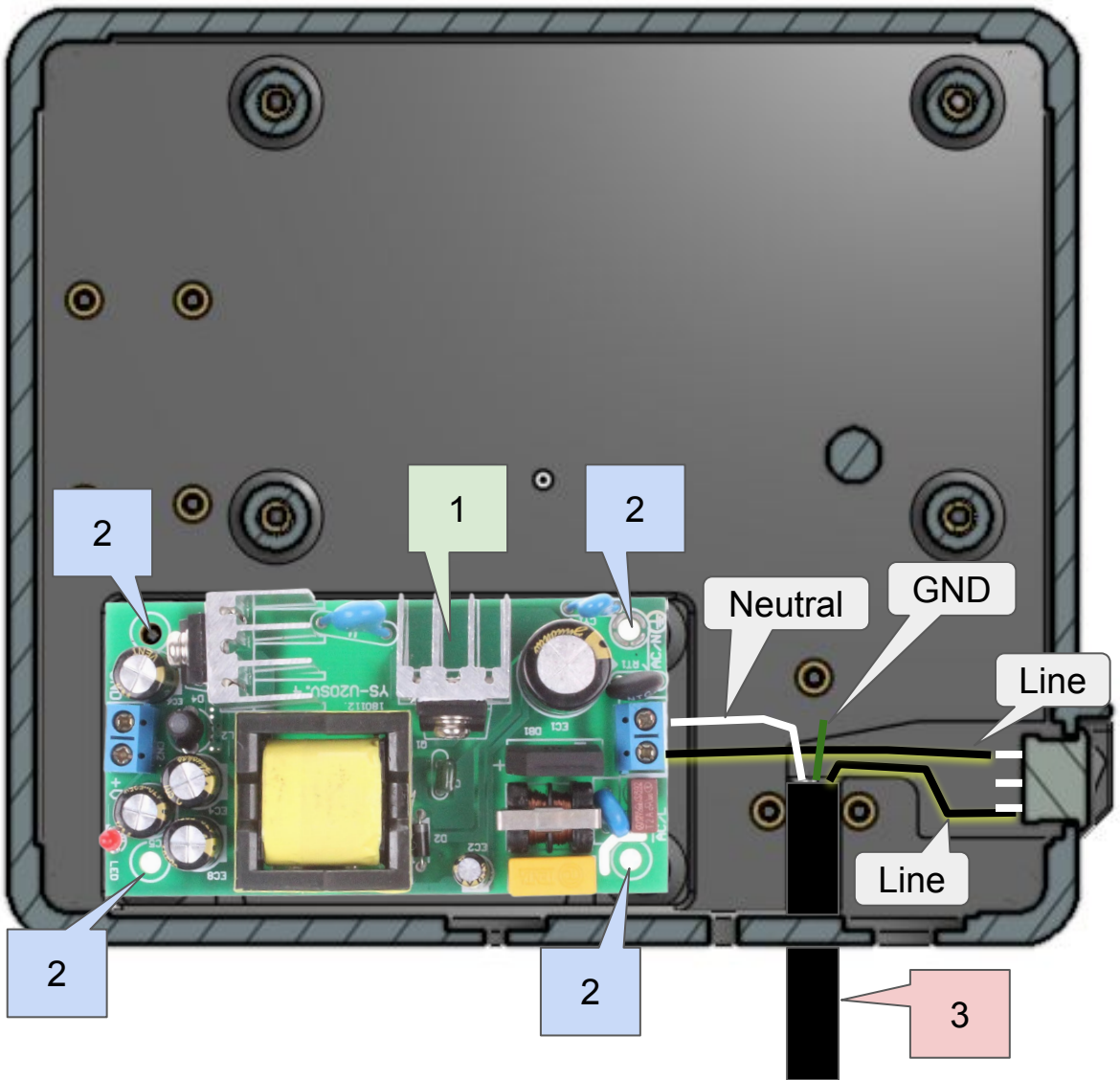
Route [Line] From the AC cord through the switch and into the [Line-In] terminal of the NOYITO AC-DC adapter

Route [Neutral] From the AC cord to the [N] Terminal on the AC adapter

Cut [GND], it isn't needed

Install Cable Clamp with with the following hardware:

4. SHCS, M3x0.50, 6mm [QTY. 2]



Install Axworthy PCB

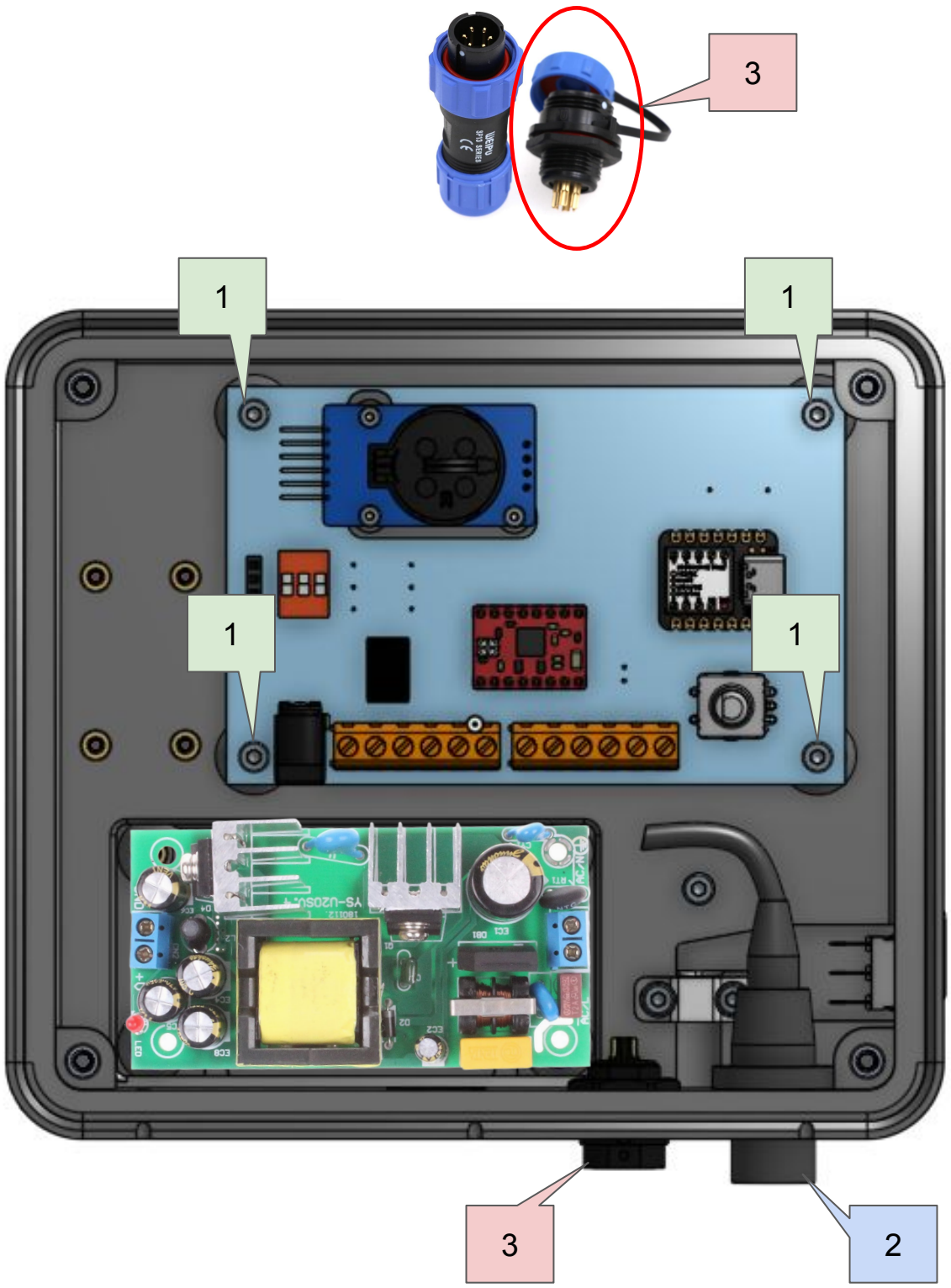
Install Axworthy PCB using the following hardware:

1. SHCS, M3x0.50, 6mm [QTY. 4]

Install The following Connectors:

2. USB-C panel-mount cable (flush/dash mount)
3. WEIPU SP1312 rear-nut

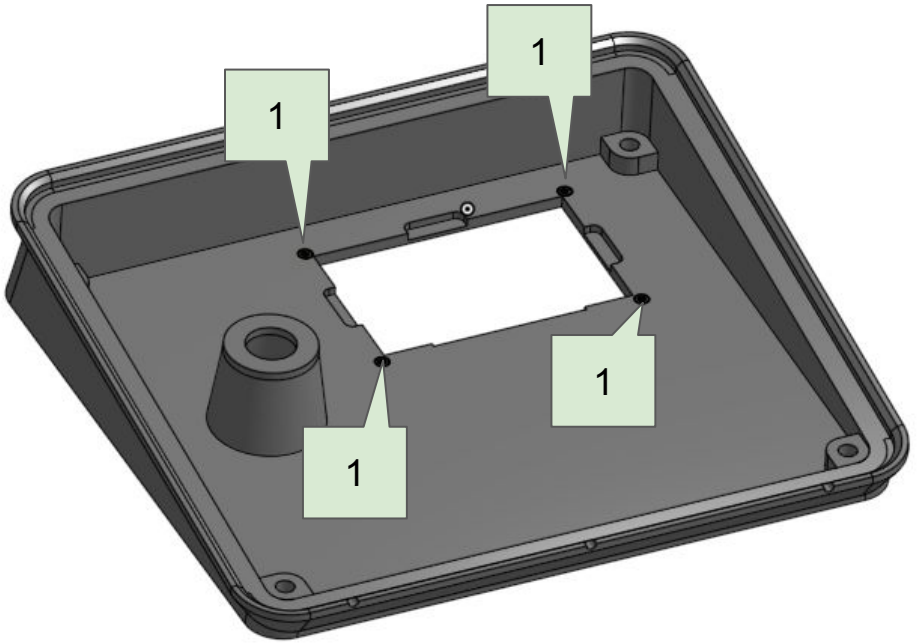
Wire PCB to AC/DC Adapter and WEIPU SP13 Per Interconnect Diagram



Lid Subassembly

Install the following heat-set inserts:

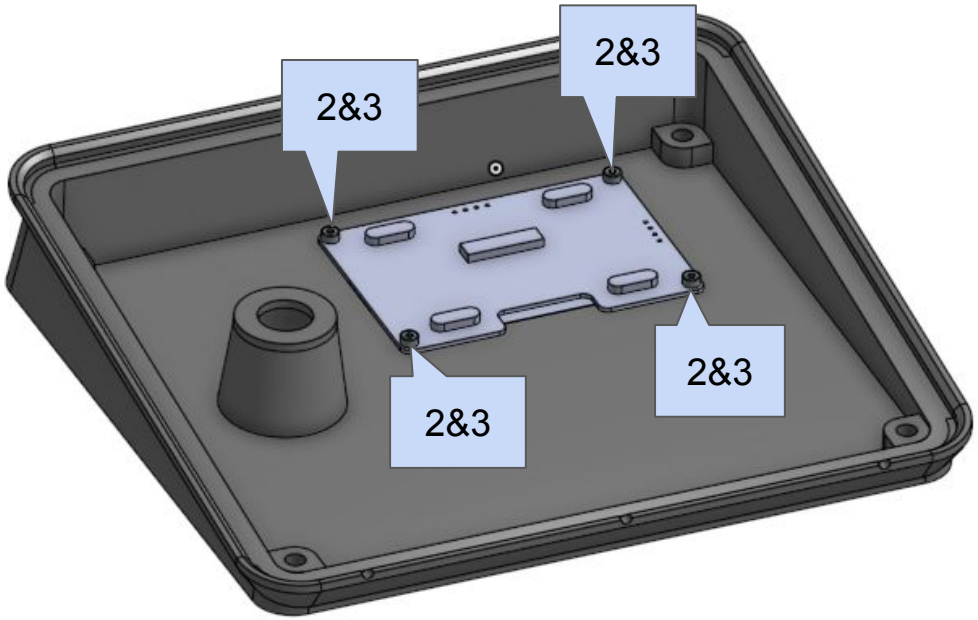
1. Heat-set insert, M2 [94180A307] [QTY. 4]



Install SSD1309 2.42" OLED (I2C) with the following hardware:

2. SHCS, M2x0.40, 10mm [QTY. 4]
3. M2 Washers [4 to 8]*

NOTE: Use M2 washers as necessary to back the end of the screw from the opposite face. The end of the M2 screw must not protrude from the other end.



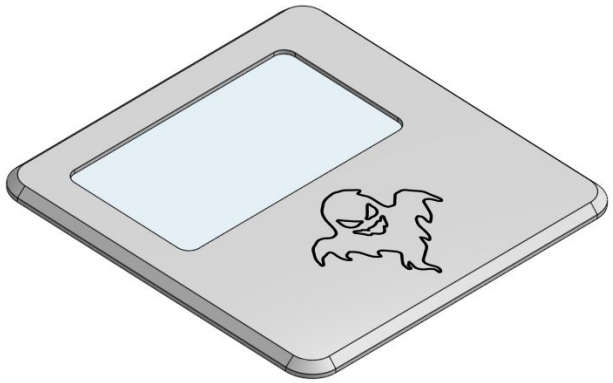
Screen Guard Subassembly

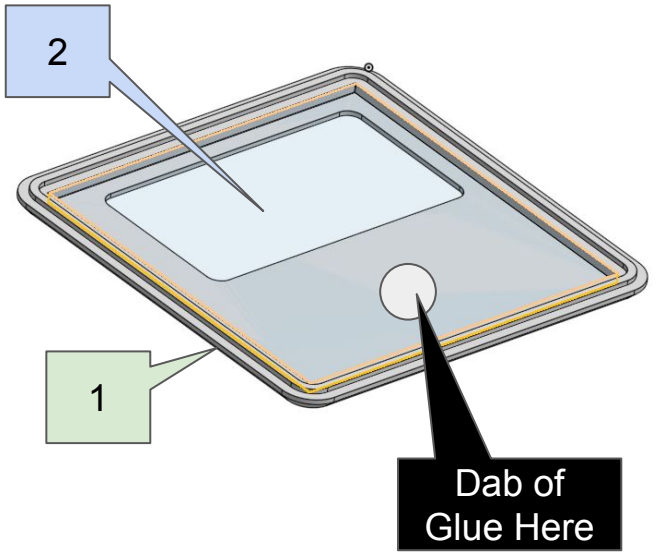
Apply a small amount of super glue between the following components:

<u>Caution</u>: Remove protective film from of Acrylic Sheet first!

1. 3x3 Acrylic Sheet
2. Screen Guard

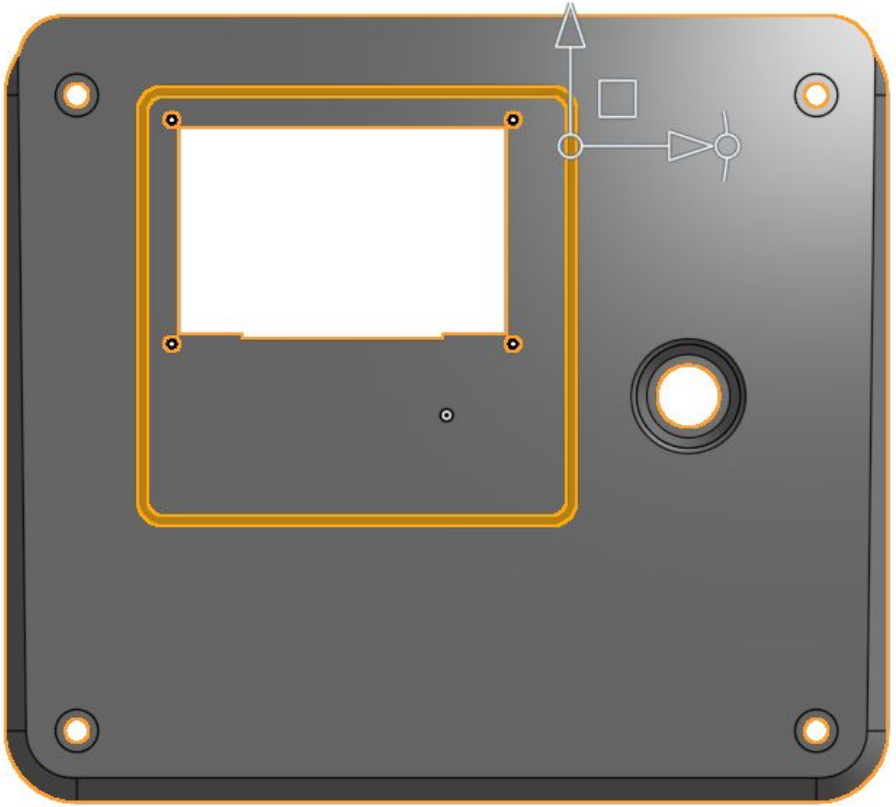
NOTE: Use just enough glue to lightly secure the sheet to the screen guard so you can hold it upside down without falling for the next step.





Install Screen Guard

Apply Super Glue to the Screen Guard groove (Highlighted in Orange)



<u>Caution:</u>

- *Don’t apply too much, or it could spill out elsewhere*
- *Apply the glue evenly to ensure connection is water tight*
- *It is highly recommended you temporarily remove the LCD before this step*

Apply Screen Guard Subassembly to groove with glue. Hold firmly for 1 minute until glue has adhered.



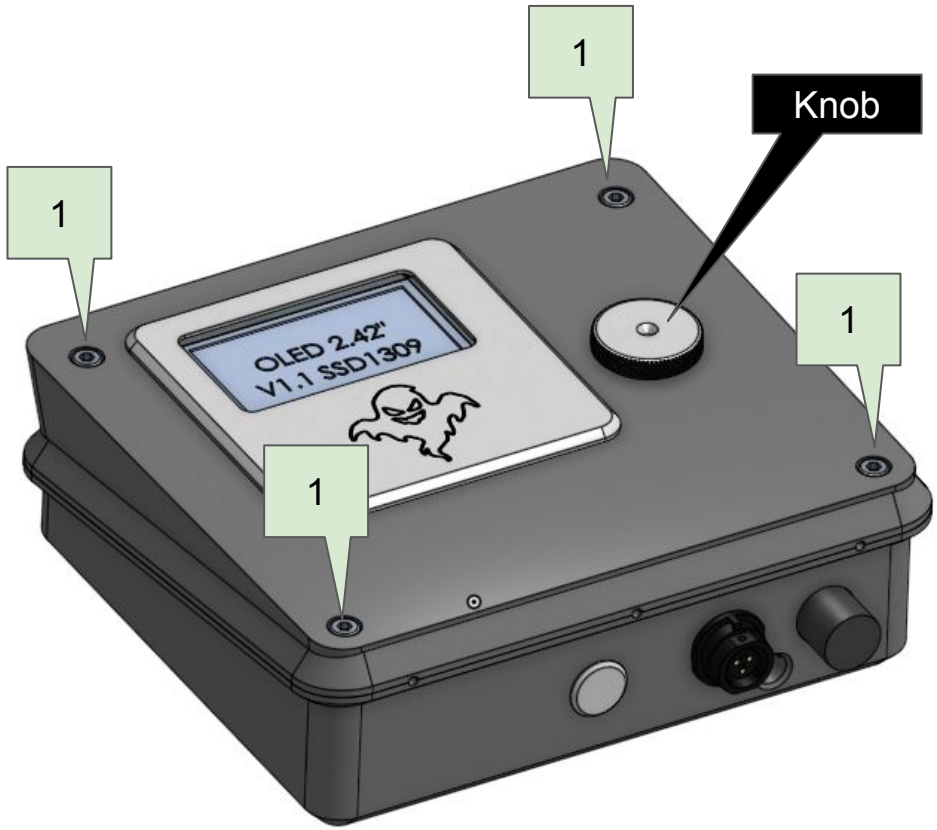
Reinstall LCD per Lid assembly steps

Install Lid to Box

Wire SSD1309 2.42" OLED (I2C) per Interconnect Diagram

Place Lid over box and fasten with the following hardware:

1. SHCS, M4x0.70, 30mm [QTY. 4]



Press Knob onto Potentiometer Shaft

NOTE: The knob has a flat side to of the shaft that needs to align with the flat face of the potentiometer shaft.

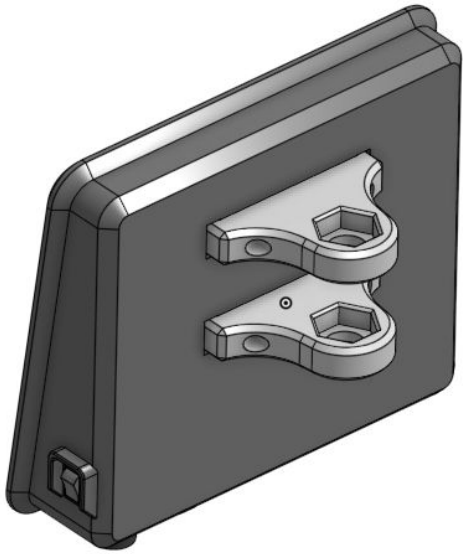
Insert Hole Plug (For AC Option only)

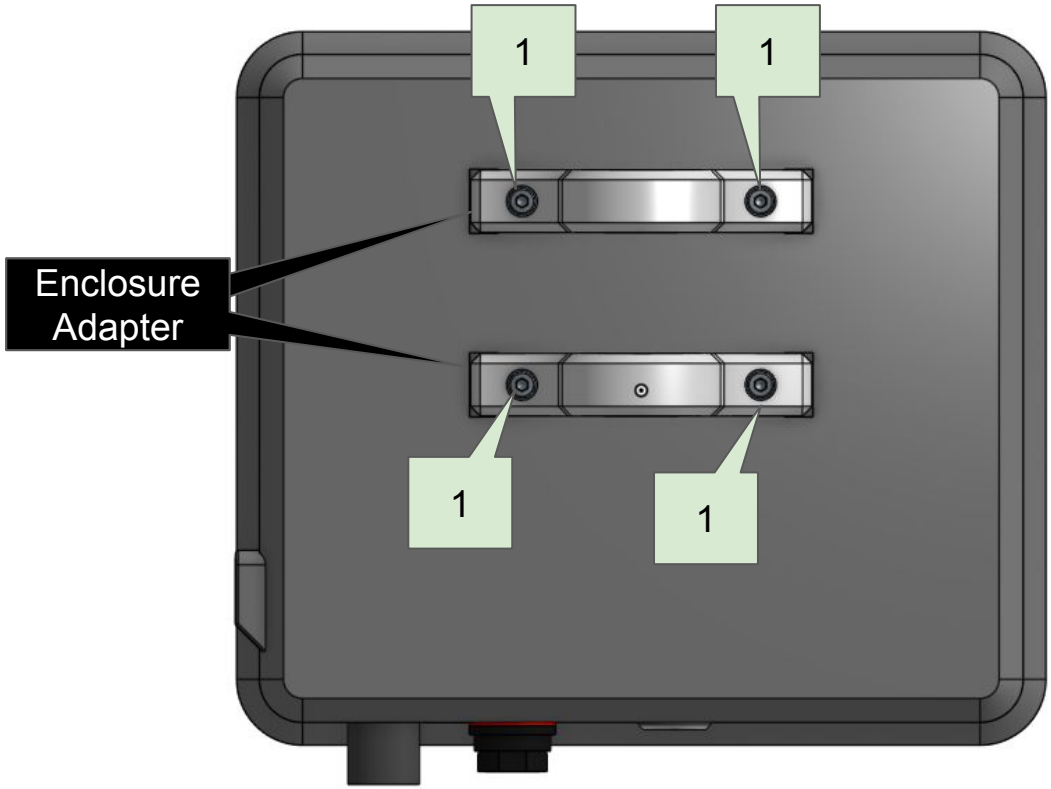
Install Enclosure Adapters

Install Enclosure Adapters to the back of the Control box with the following hardware:

1. SHCS, M3x0.50, 6mm [QTY. 4]

NOTE: Hex pockets on adapters need to face up





Solder Wire Harness

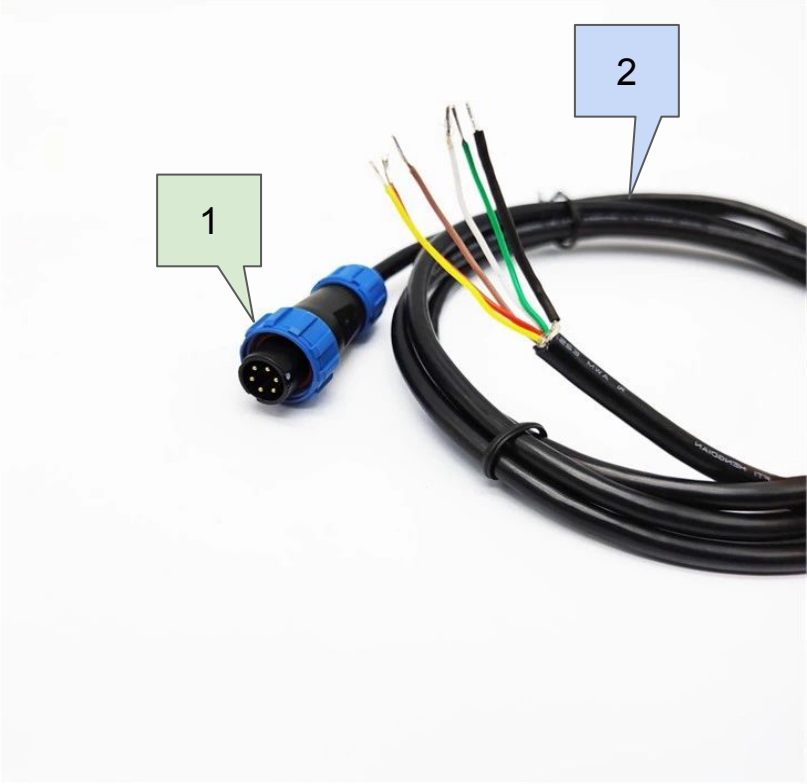
Use the following hardware to create a double ended cable harness:

1. WEIPU SP13 6-pin Cable Connector [QTY. 2]
2. 22/6 stranded cable (tinned) [QTY. 1]



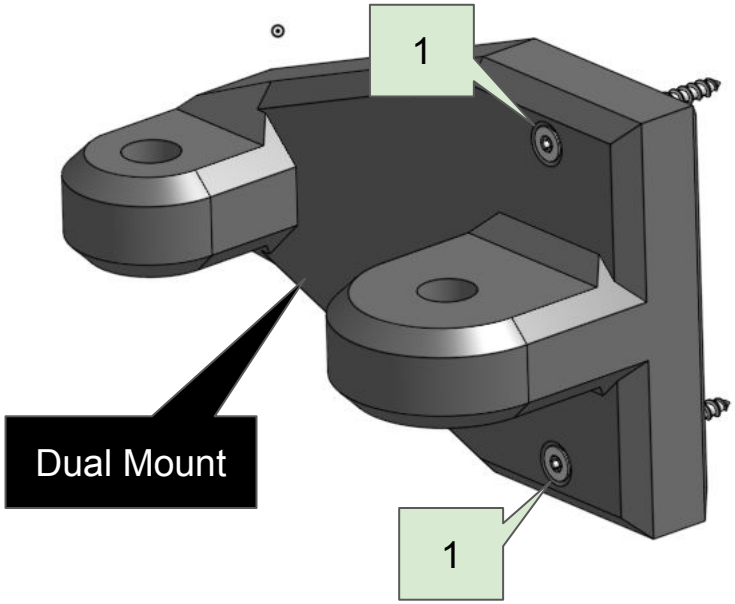
See Interconnect Diagram (AC or DC) for soldering details

Connect Harness this between the Controller and Drive Housing



Mounting & Stringing

Mounting Option 1: Dual Mount

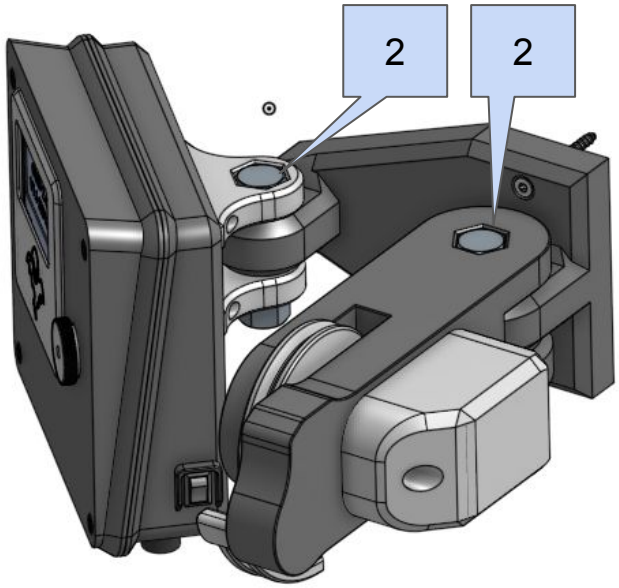


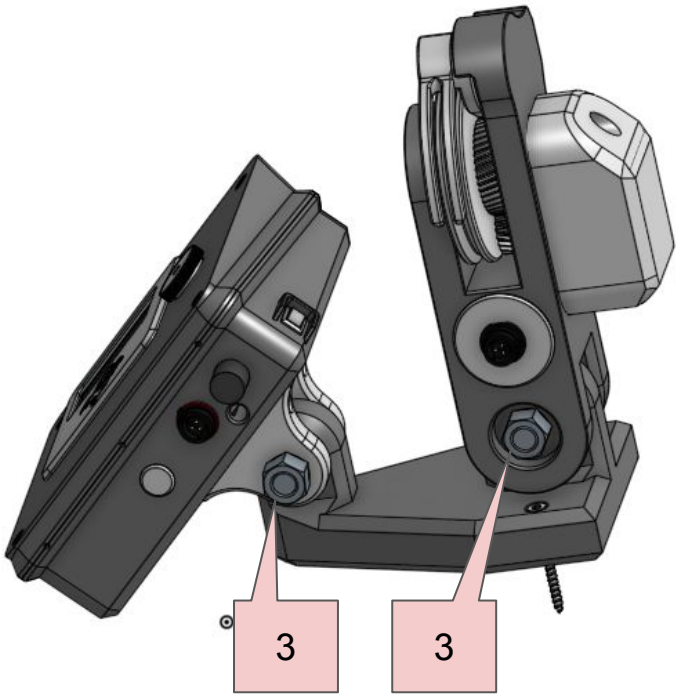
Secure Dual Mount to solid wood surface using the following Hardware:

1. Wood Screw, #8 x 2” [QTY. 2]

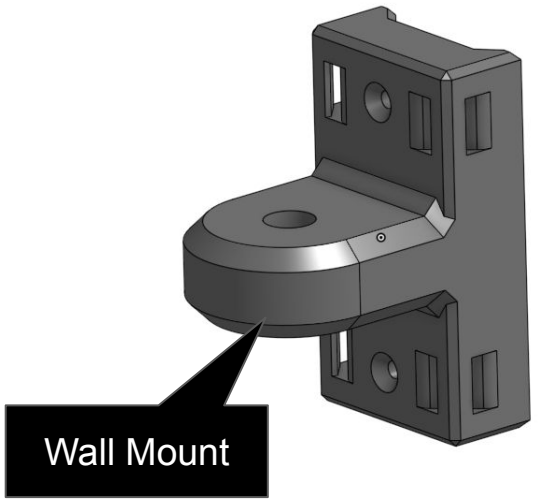
Fasten Controller and Drive Assembly with the following hardware:

2. Hex head bolt, M12x1.75, 55mm [QTY. 2]
3. Hex nut, M12x1.75 (grade A&B) [QTY. 2]





Mounting Option 2: Individual Mount

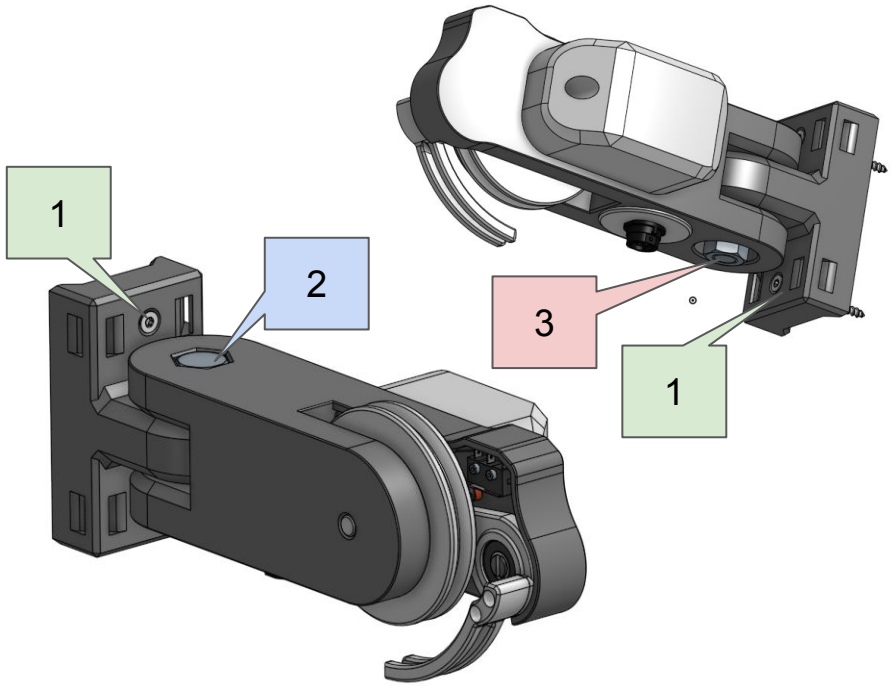


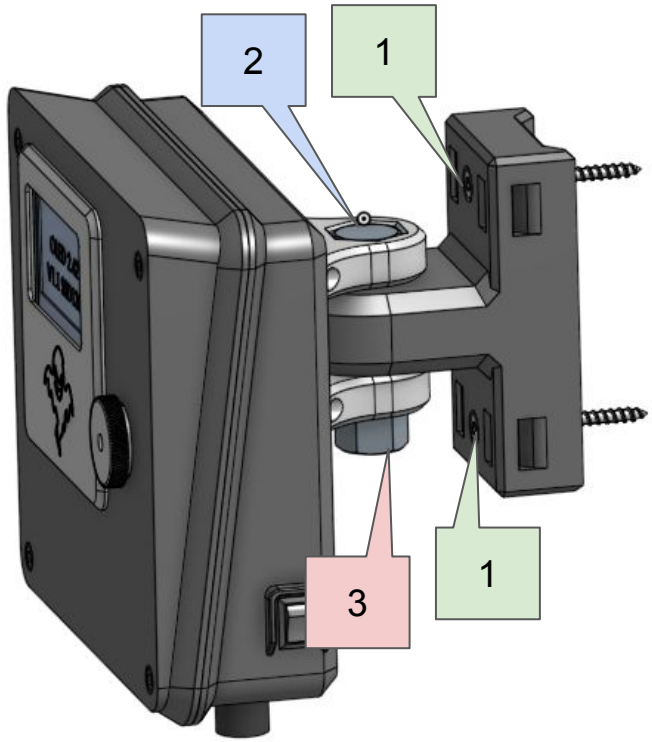
Secure 2 Wall Mounts to solid wood surface using the following Hardware:

1. Wood Screw, #8 x 2” [QTY. 4]

Fasten Controller and Drive Assembly with the following hardware:

2. Hex head bolt, M12x1.75, 55mm [QTY. 2]
3. Hex nut, M12x1.75 (grade A&B) [QTY. 2]





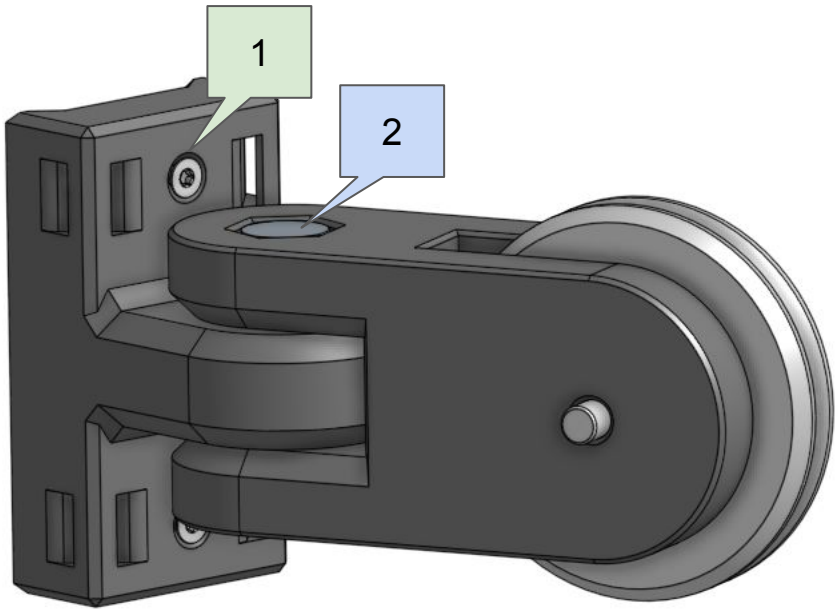
Idler Mount

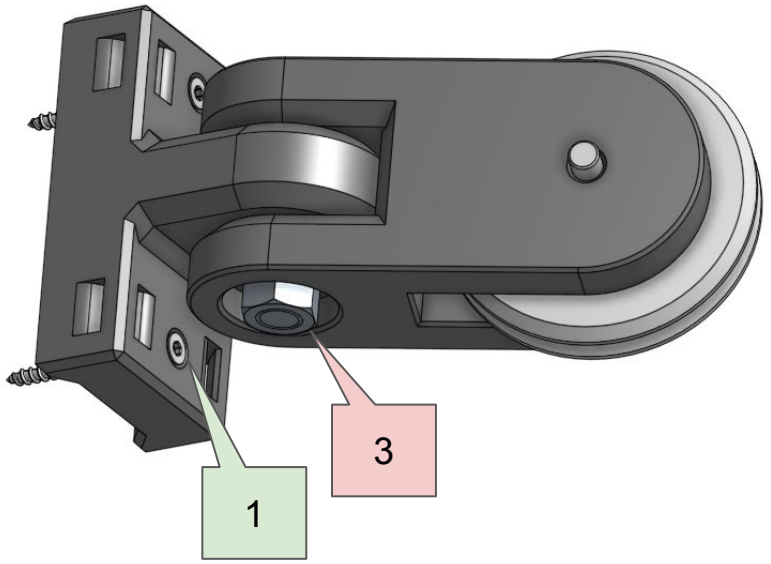
Secure Wall Mount to solid wood surface using the following Hardware:

1. Wood Screw, #8 x 2” [QTY. 2]

Fasten Idler Assembly with the following hardware:

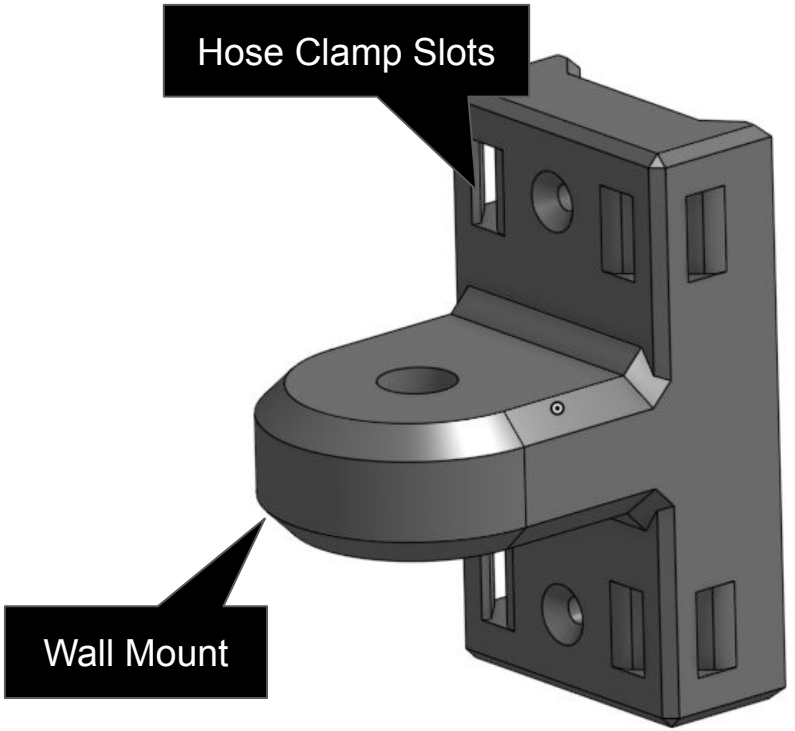
2. Hex head bolt, M12x1.75, 55mm [QTY. 1]
3. Hex nut, M12x1.75 (grade A&B) [QTY. 1]





Tree/Pole Mounting Option

Substitute Hose Clamps for Screws to fasten the Wall mount to circular objects. Select the appropriate size for your application at your local hardware store.









Solder Wire Harness

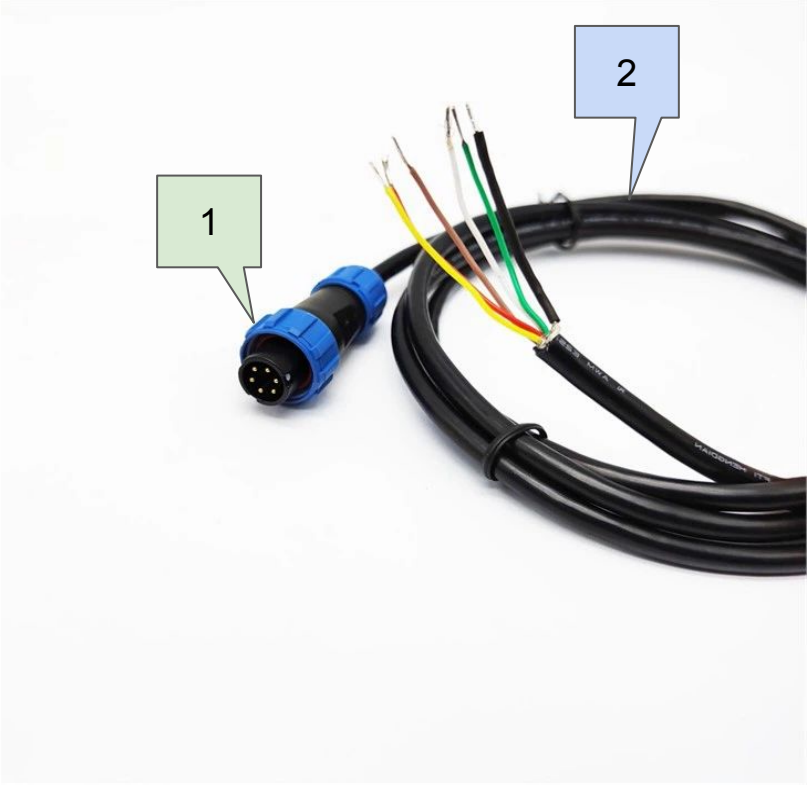


Use the following hardware to create a double ended cable harness:

1. WEIPU SP13 6-pin Cable Connector [QTY. 2]
2. 22/6 stranded cable (tinned) [QTY. 1]

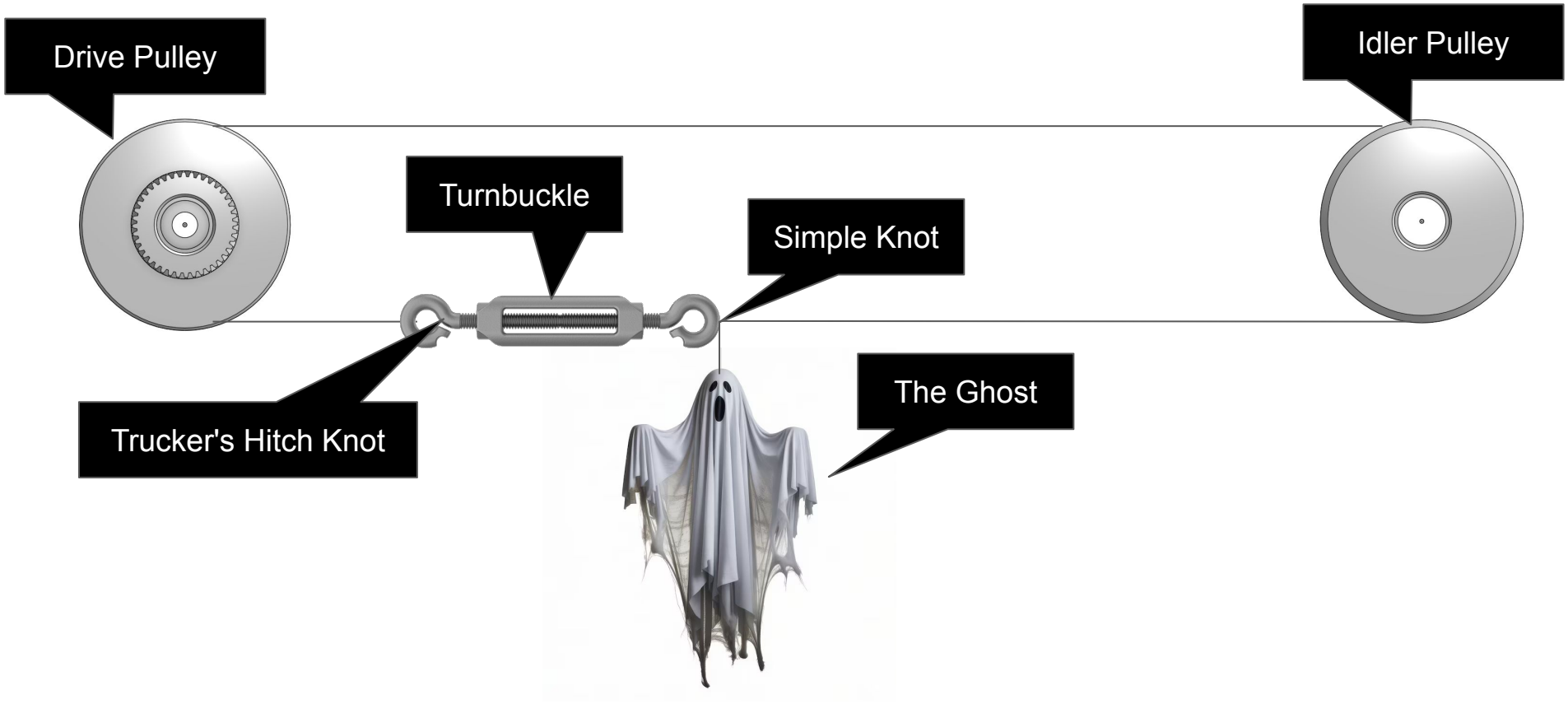
See Interconnect Diagram (AC or DC) for soldering details

Visit for connector details:
https://www.weipuconnector.com/products/sp13/



Connect Harness this between the Controller and Drive Housing

Tying the Knot



1. Route the line around both the Drive and Idler pulleys
2. Tie a simple knot to one end of the turnbuckle (Turnbuckle optional if you're a pro with knots)
3. Tie a Trucker's Hitch Knot to the other end of the turnbuckle
4. Turn your turnbuckle to tighten the line further
   a. Get the line as tight as you can without breaking it
   b. Once tight, the line should make a low frequency vibration when you strum it.
5. Hang your ghost. I like to use one of the turnbuckle hooks to loop the ghost on for easy removal later
6. Enjoy!